



《CAD与工程制图》

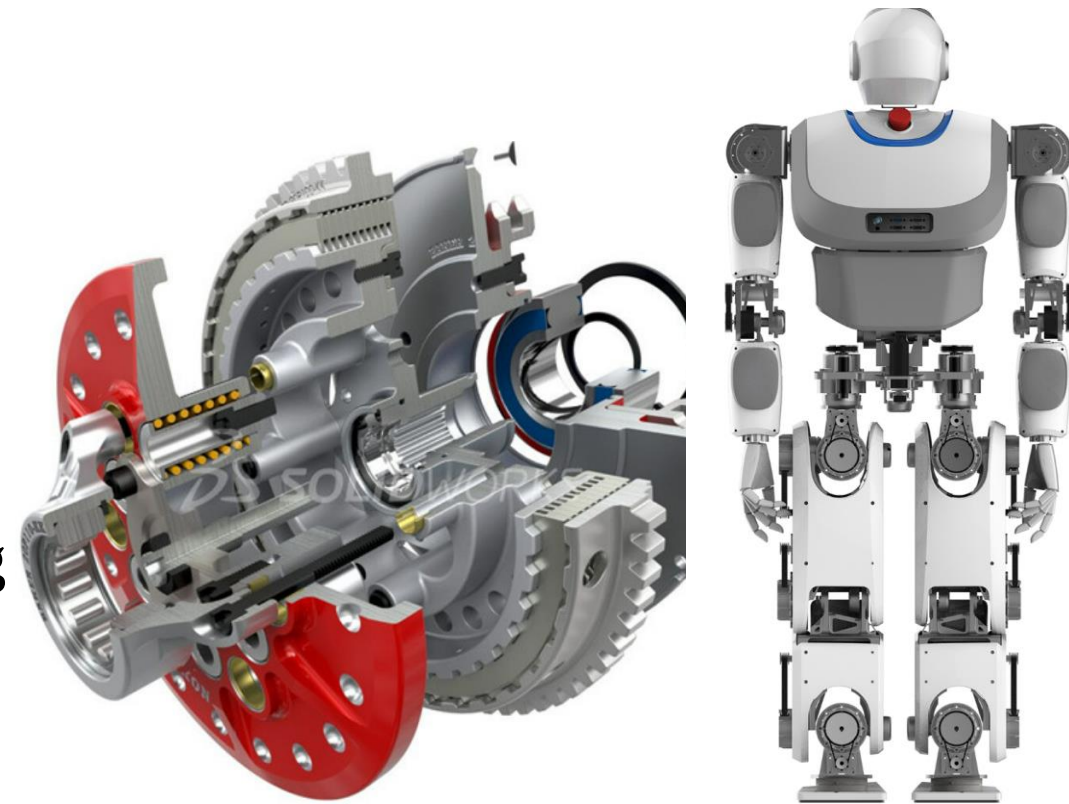
CAD (Computer Aided Design) and Engineering Drawing

2024年 秋季

Yixuan Guo 郭艺璇

Department of Mechanical and Energy Engineering

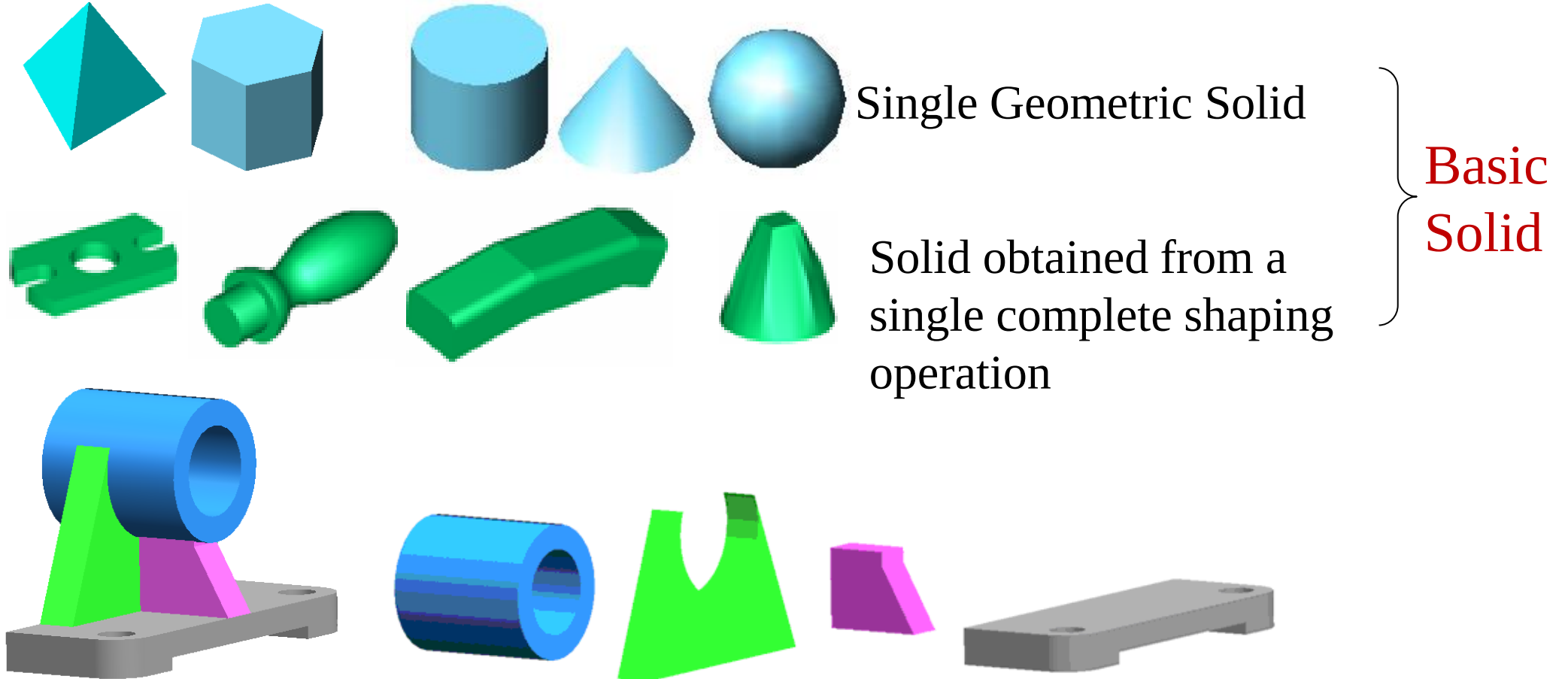
Southern University of Science and Technology



Chapter 5 Assemblies and Their Design

- **5-1 Shape Analysis and Combination Forms of Assemblies**
- **5-2 Drawing of Intersection Lines**
- **5-3 Drawing of Assemblies**
- **5-4 Dimensioning of Assemblies**
- **5-5 Reading Drawings of Assemblies**
- **5-6 Shape Design of Assemblies**

Review: Classification of Geometric Solids



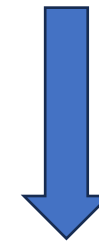
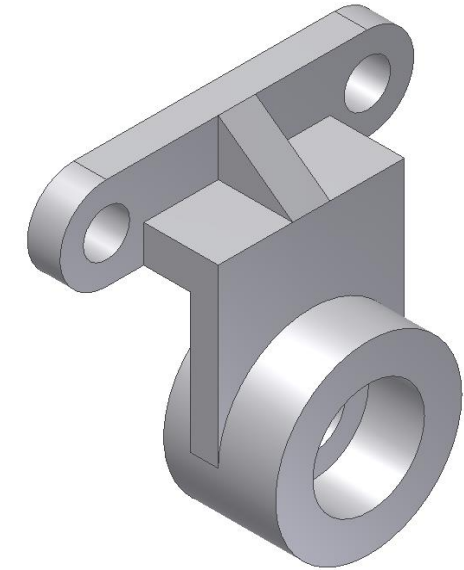
Assembly: A complex shape formed by the organic combination of **several basic solids** according to certain relative positions and combination methods.

I. Shape Analysis of Assemblies

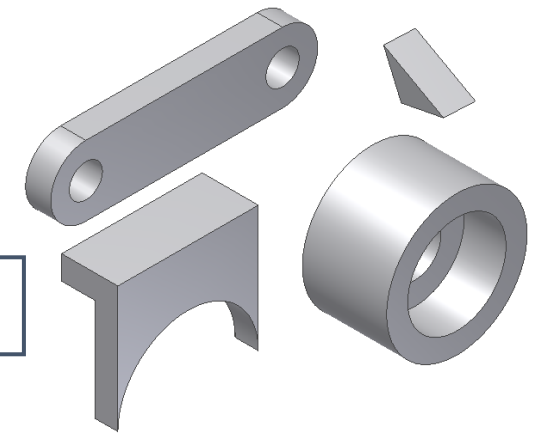
Shape Analysis: A method of analyzing a complex object by hypothetically decomposing it into **several basic solids** based on shape characteristics.

Shape analysis is the basis for **drawing, reading drawings, and dimensioning**.

Shape analysis method involves **hypothetically decomposing** an assembly to facilitate the analysis of the drawing. It is not an actual decomposition of the assembly. When **drawing, reading drawings, and dimensioning**, the object should still be considered as a **whole**.



Consider:
Why is it divided in this way?



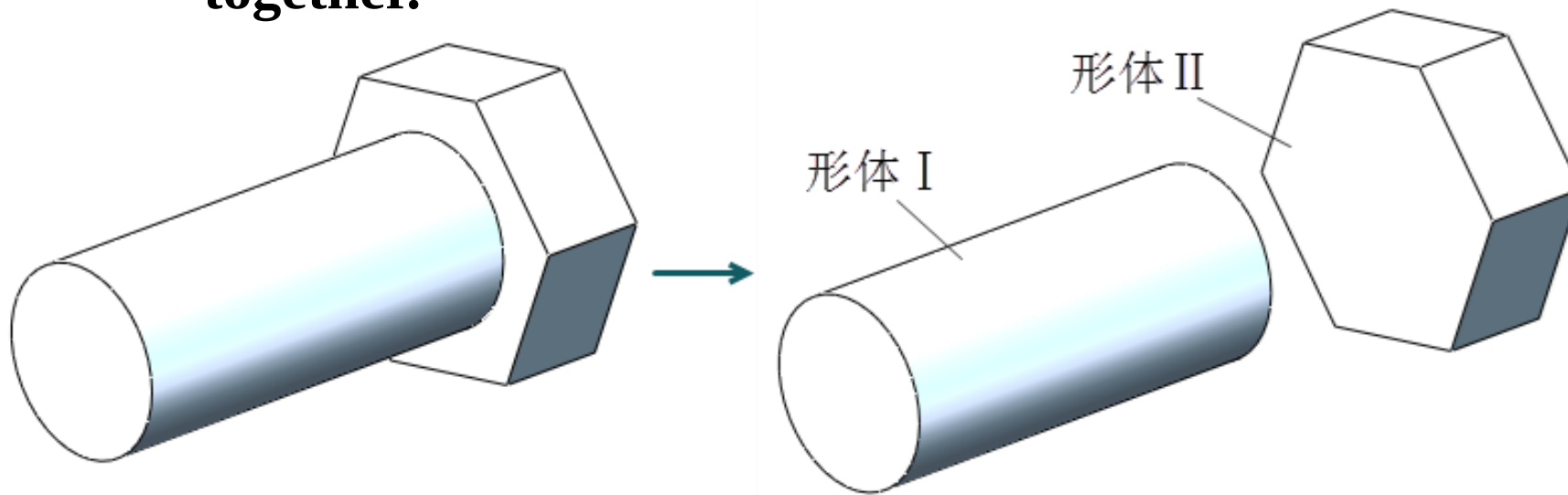
II. Combination Forms of Assemblies

Assemblies:

A solid formed by the **superposition** or **cutting** of basic solids is called an assembly.

Superposition Assemblies:

An assembly of superposition is formed by stacking basic solids together.



Superposition Assemblies

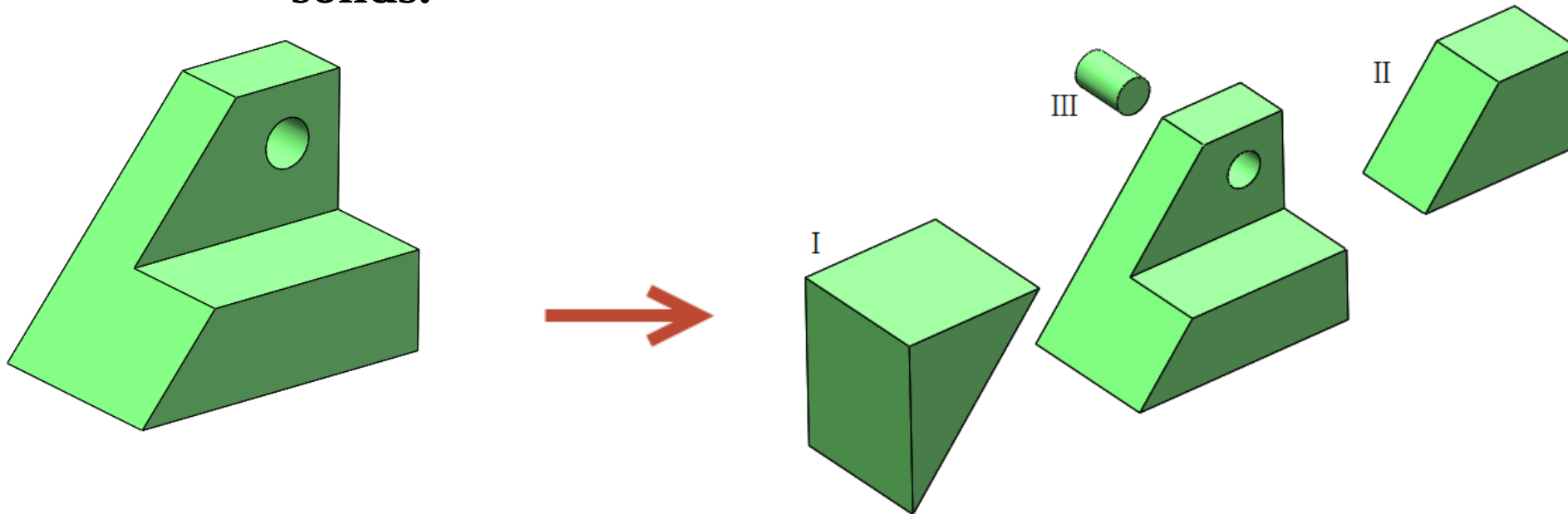
II. Combination Forms of Assemblies

Assemblies:

A solid formed by the **superposition** or **cutting** of basic solids is called an assembly.

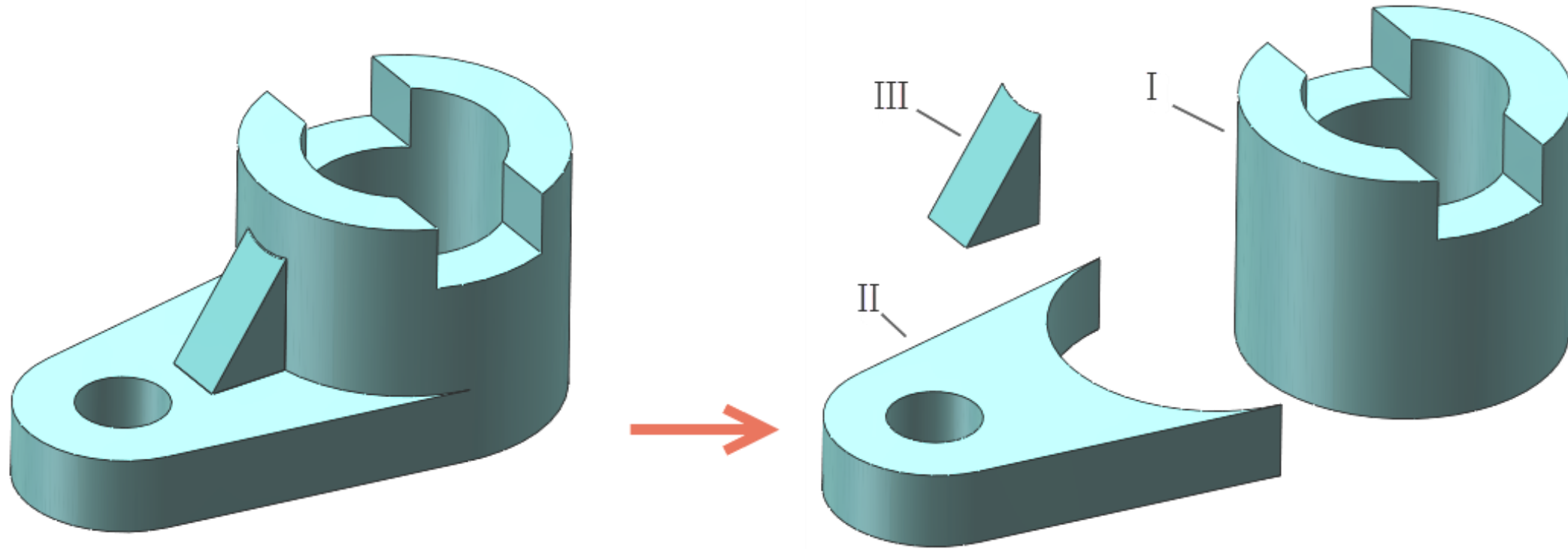
Cutting-style Assemblies:

A cutting-style assembly is formed by cutting and excavating basic solids.



Cutting-style Assemblies

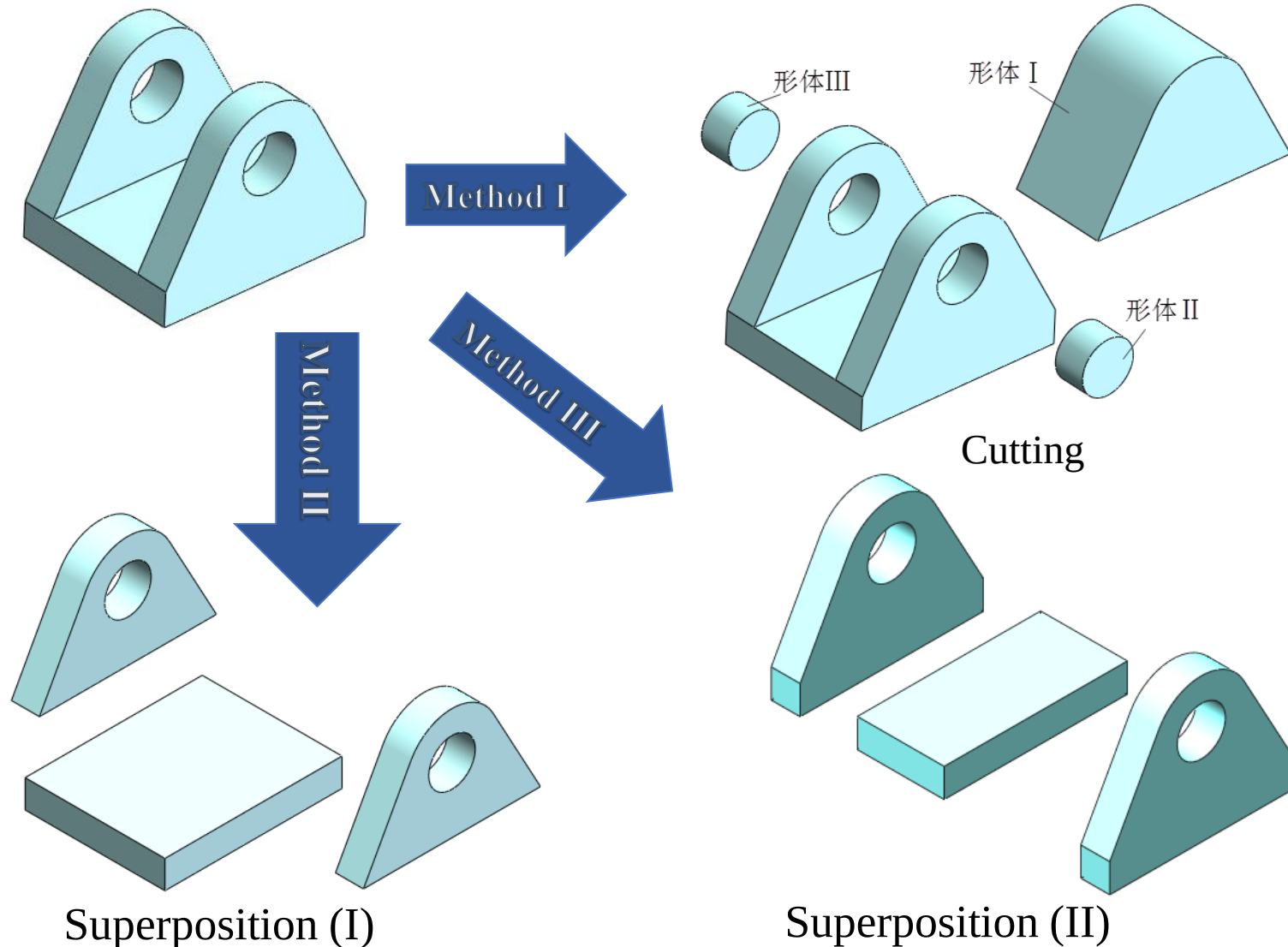
II. Combination Forms of Assemblies



In general, in an assembly, both forms often coexist, with **both superposition and cutting** (excavation).

II. Combination Forms of Assemblies

The formation
method of the
same assembly is
not unique.

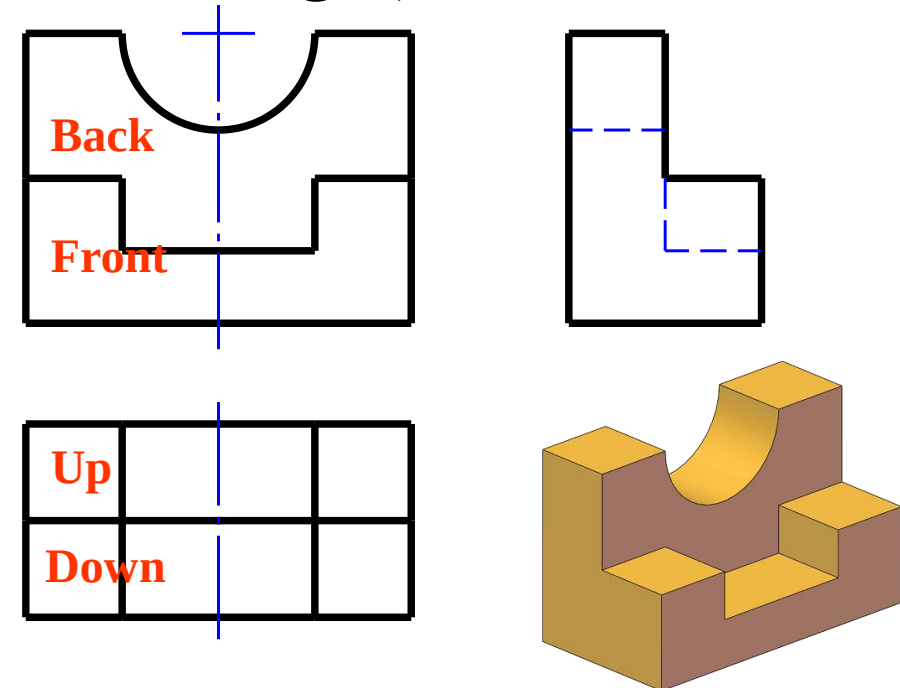
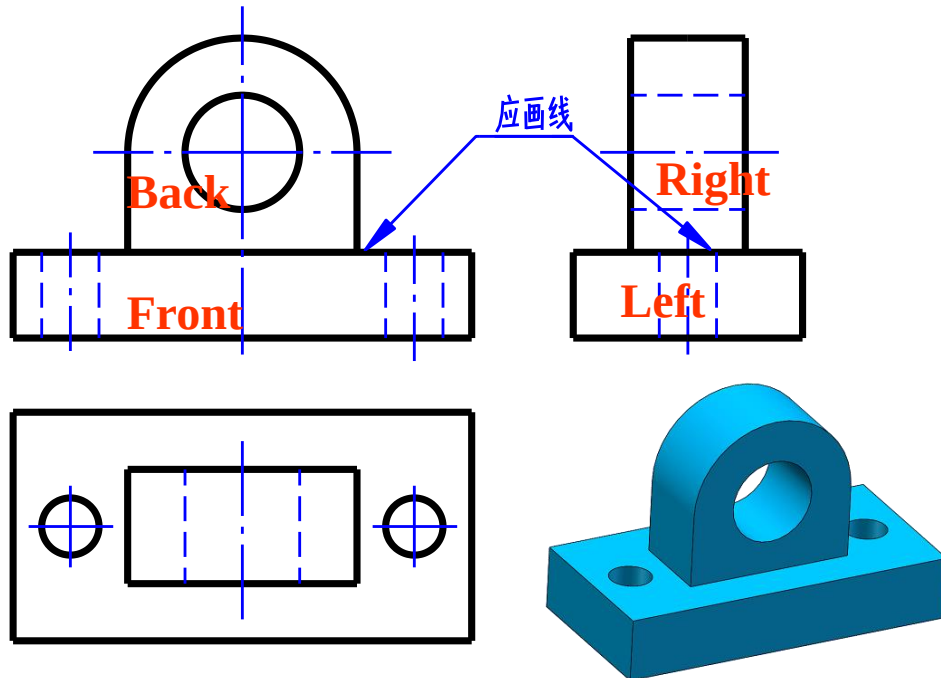


III. Surface Connection Relationships of Assemblies

The relative positions or properties of the surfaces of an assembly are different, and so are their connection relationships. Generally, they can be divided into four types: **offset**, **coplanar**, **intersecting**, and **tangent**.

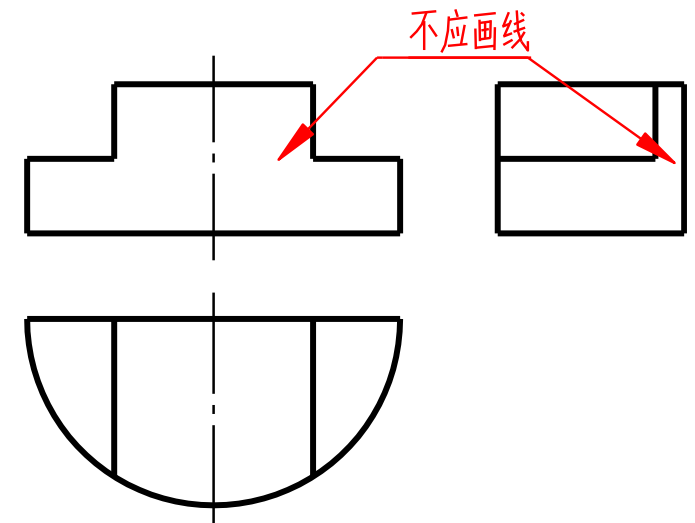
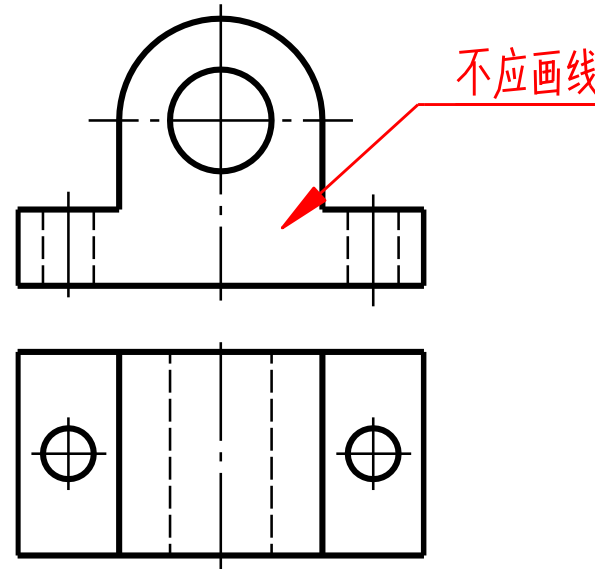
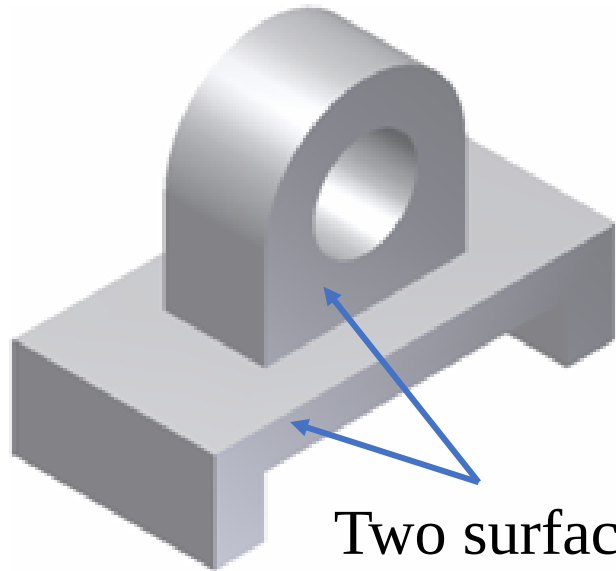
1. Offset

Offset refers to the situation where **two parallel surfaces** of two solids are **offset** from each other in the front-back (or up-down, left-right) direction.

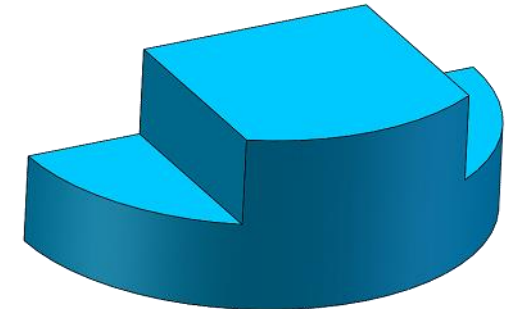
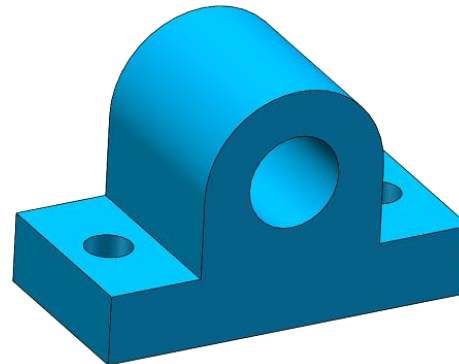


2. Coplanar

Coplanar means that some **surfaces of two solids coincide** with each other. When the surfaces of two solids are **coplanar**, **no dividing line** should be drawn in between.



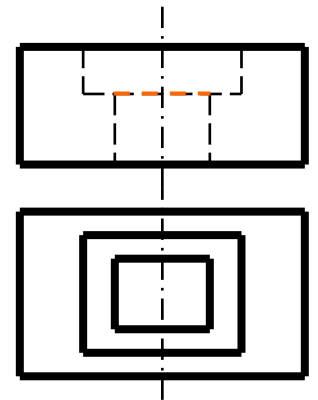
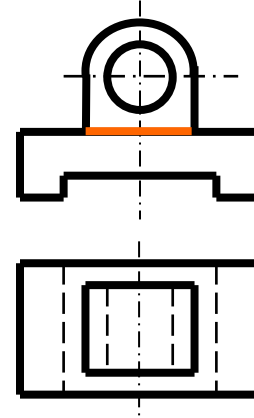
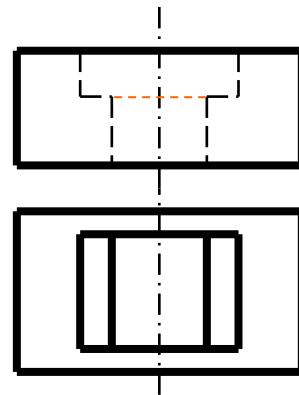
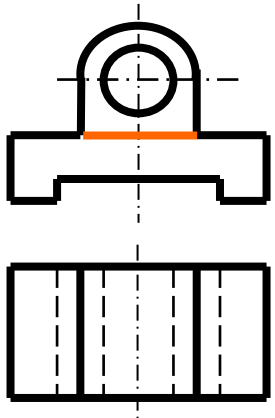
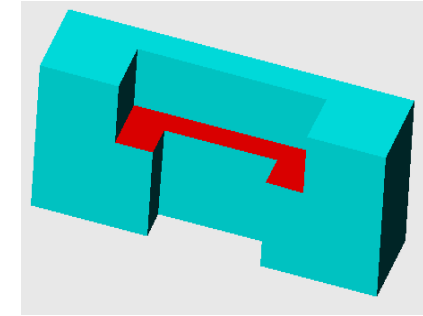
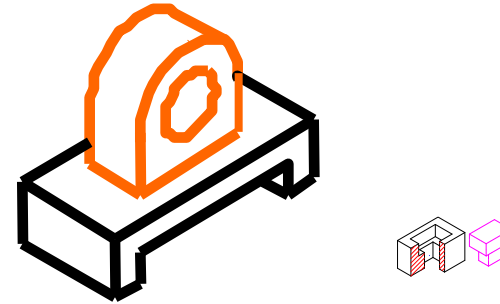
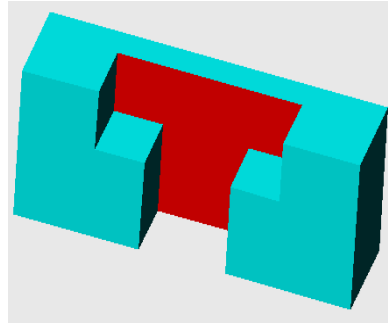
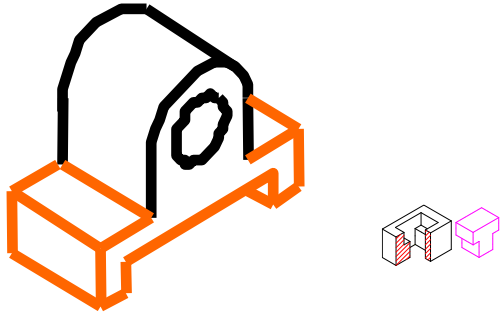
Consider:
What are the manifestations of coplanarity in 3D modeling operations?



Coplanar: No dividing line

VS

Offset: Draw dividing line



**Solid Addition
(Coplanar)**

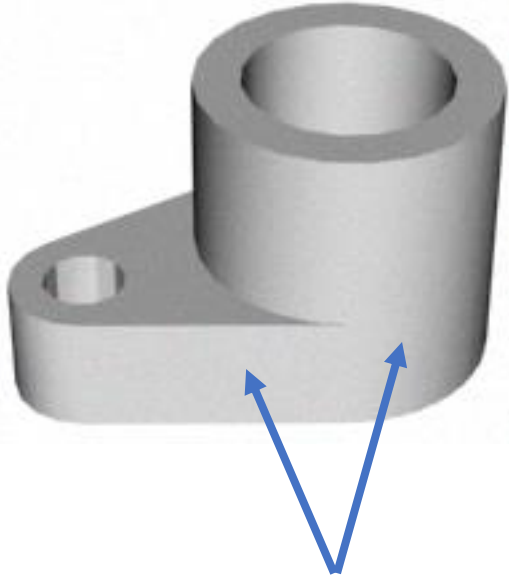
Void Addition (Coplanar)

**Solid Addition
(Offset)**

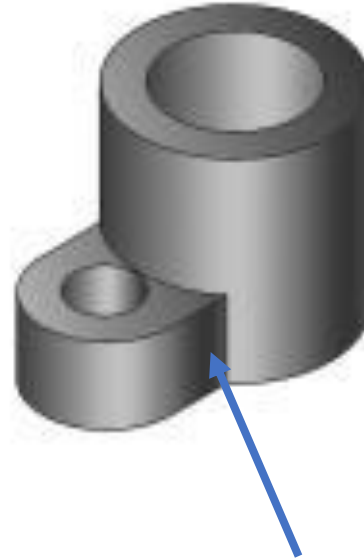
**Void Addition
(Offset)**

3. Tangent

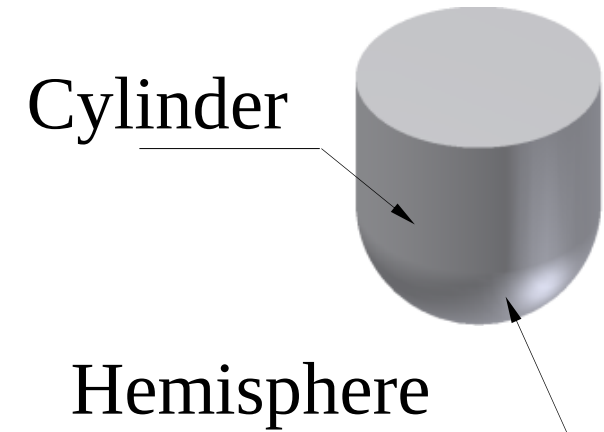
Some surfaces of two basic solids **connect smoothly** at a certain point **without a clear dividing line**.



A plane and a curved surface are **directly tangent**, with a smooth transition.



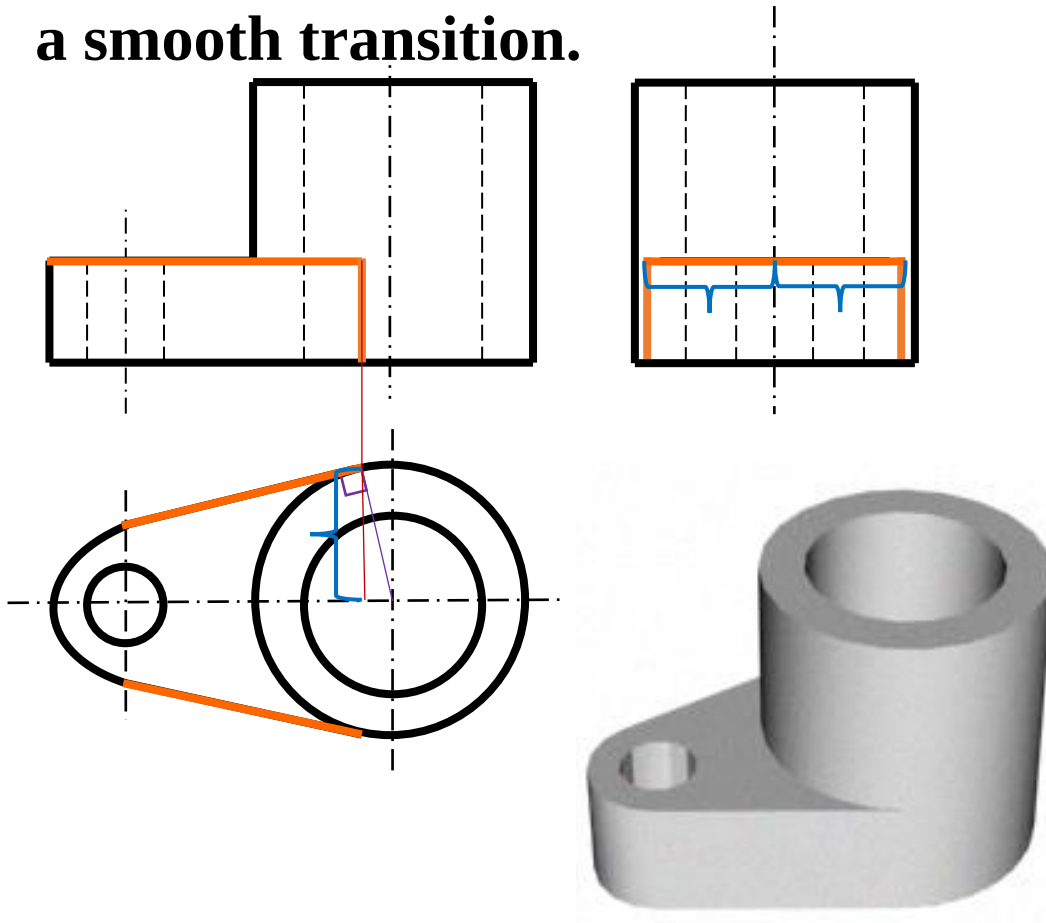
A plane and a curved surface are **tangent through a shared arc**, with a smooth transition.



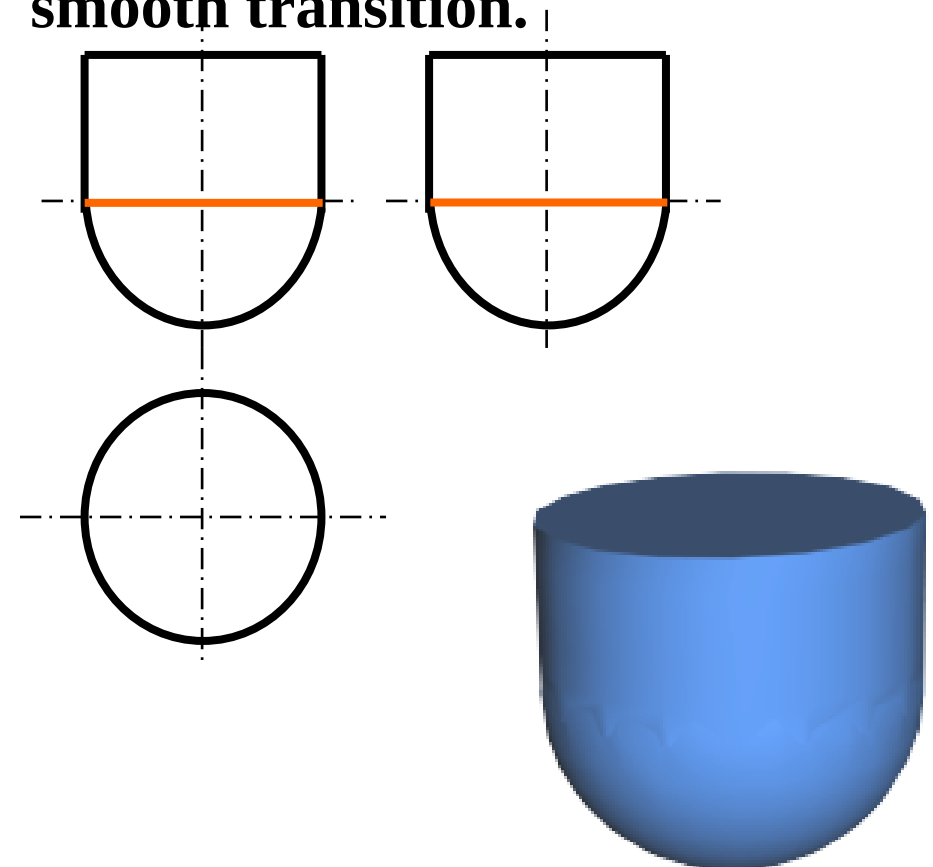
Curved surfaces are tangent to each other, with a smooth transition.

Drawing of tangency: Locate the point of tangency, no draw the intersection line.

A plane and a curved surface have a smooth transition.

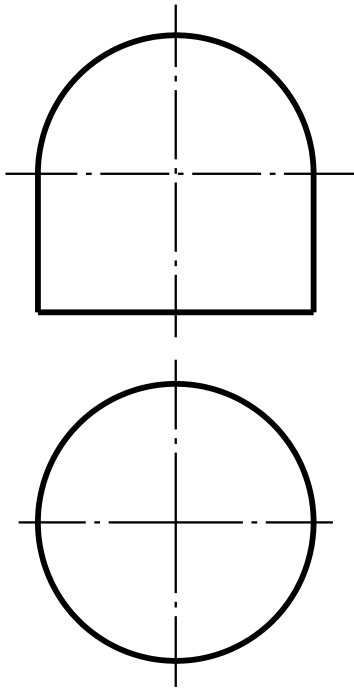


Curved surfaces have a smooth transition.

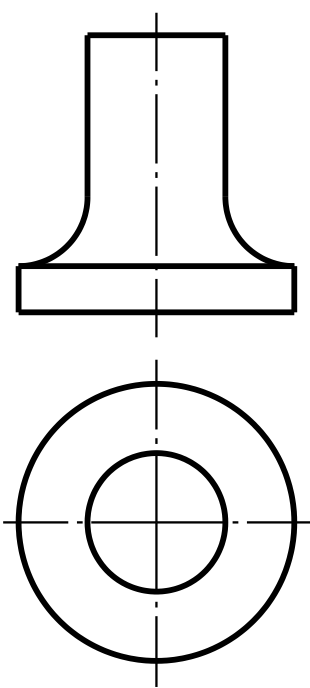


3. Tangent

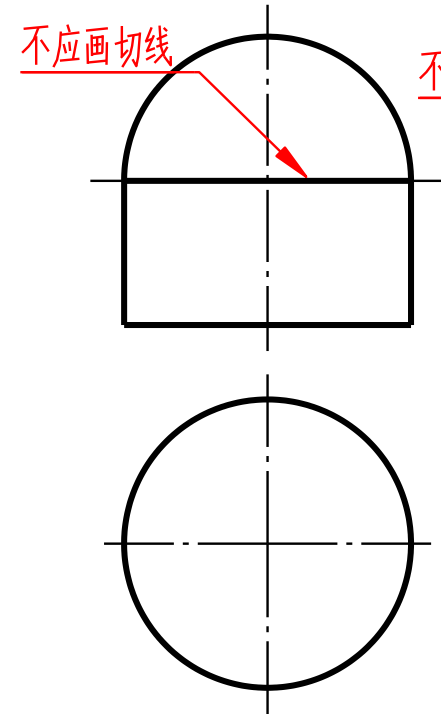
Example: Determine whether the following drawings are correct



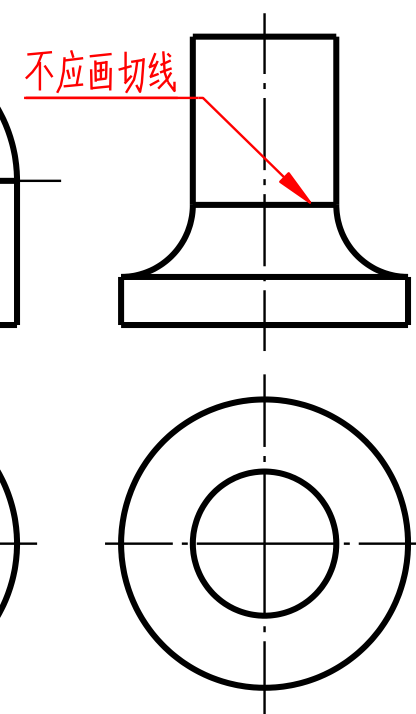
Correct



Correct



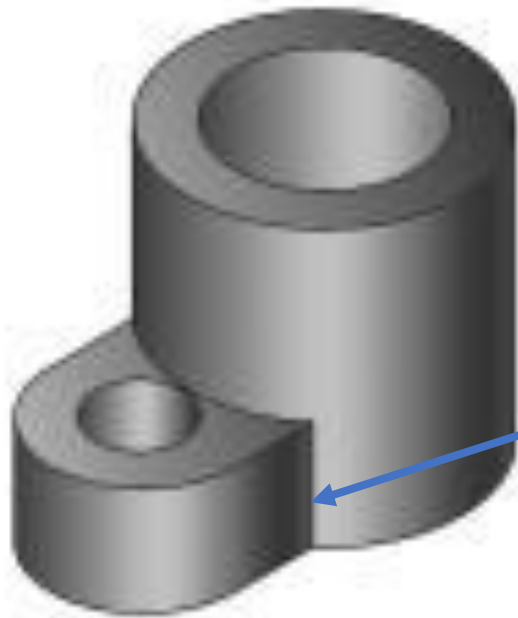
Wrong



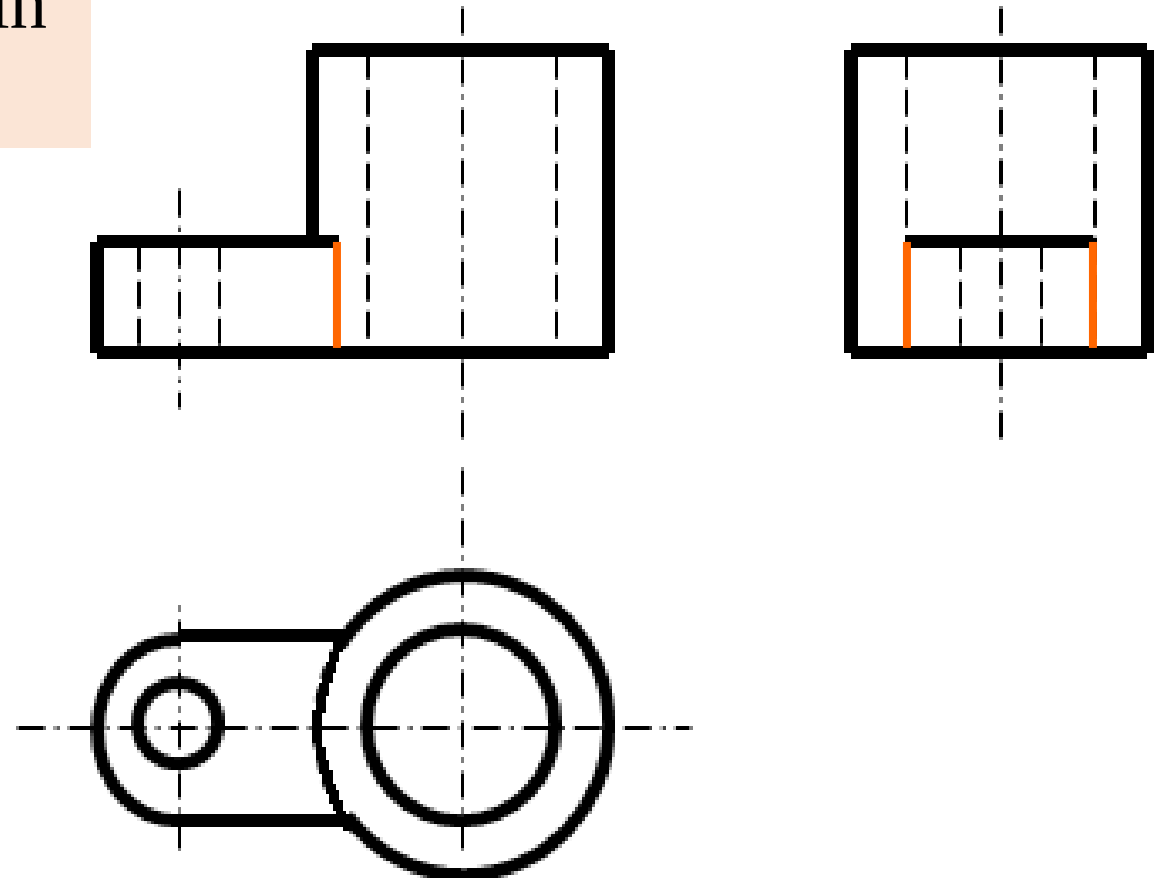
Wrong

Consider:

If it is not an arc connection, how should the intersection line of the two cylinders in the figure above be drawn?

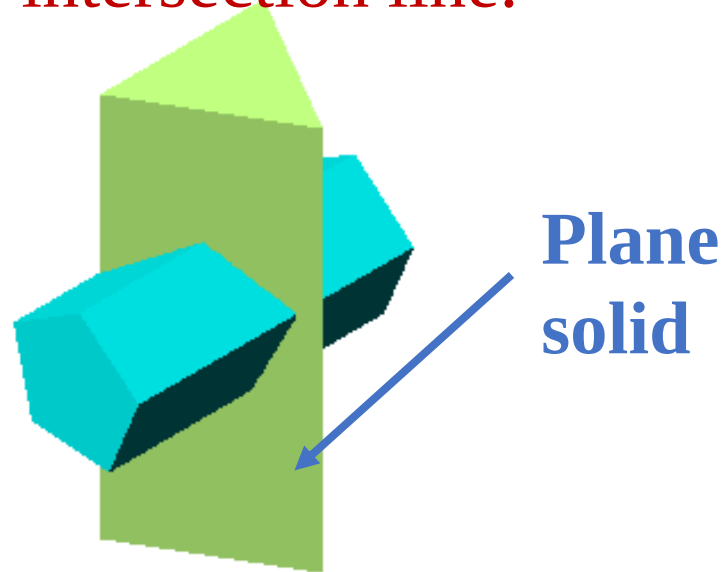


Non-smooth
transition

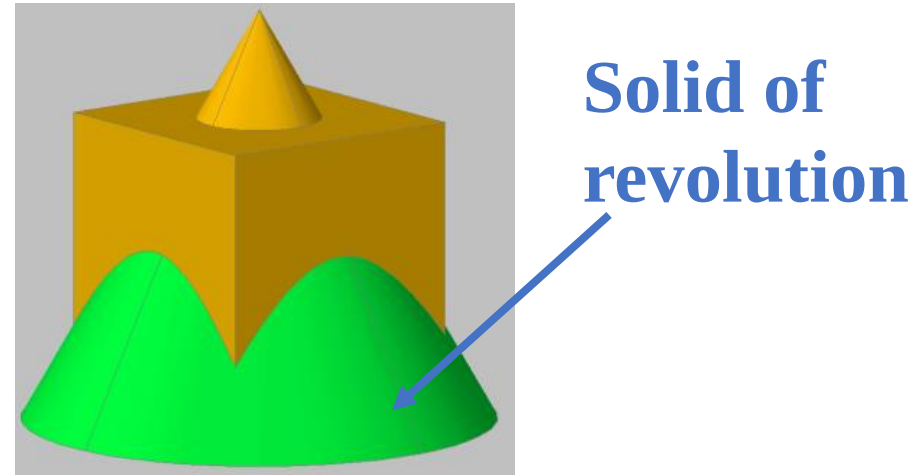


4. Intersecting

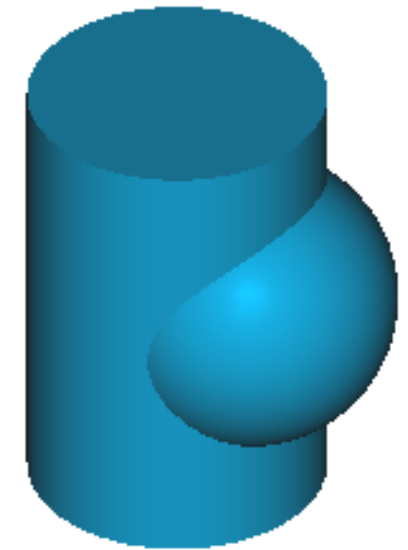
Some surfaces of two basic solids intersect, producing an intersection line.



Intersection of **plane solids**



Intersection of a **plane solid** and a **curved solid**



Intersection of **curved solids**

Chapter 5 Assemblies and Their Design

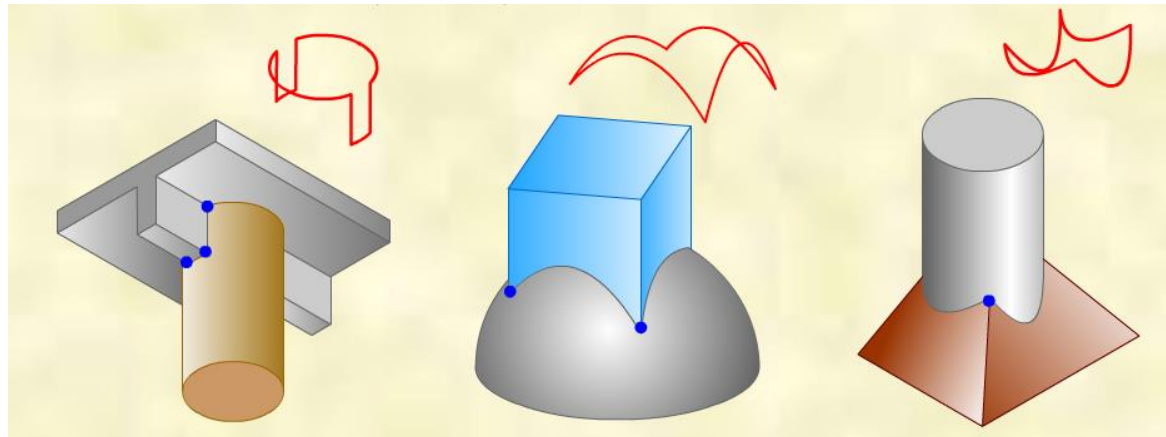
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I. The Concept of Intersection Lines

Intersection line: The line produced by the intersection of two basic solids.

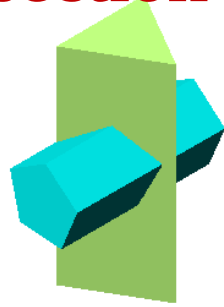
Characteristics of Intersection Lines:

1. The two solids involved in the intersection are of finite size, so the intersection line is closed.
2. The intersection line is **a line shared by the surfaces** of the two solids and also serves as **the boundary line** between them. The intersection line can also be regarded as **the set of points common to the surfaces** of the two solids.

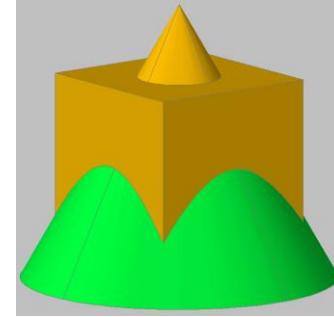


II. Classification of Intersection Lines

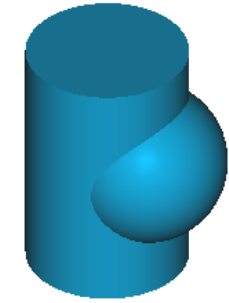
1. Classification by the types of solids involved in the intersection



Intersection of plane solids

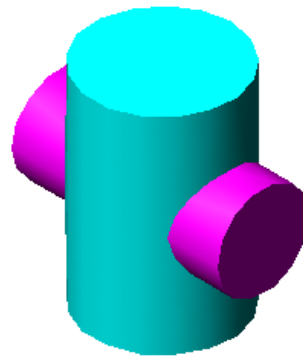


Intersection of a plane solid and a curved solid

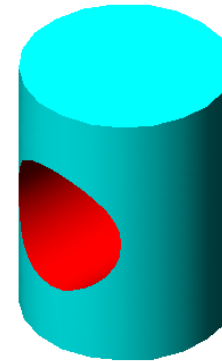


Intersection of curved solids

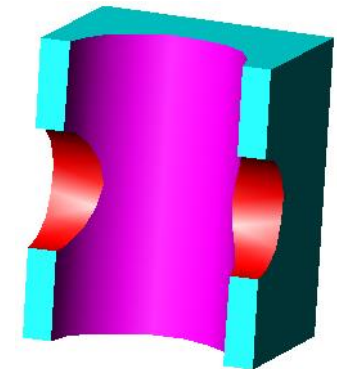
2. Classification by whether the solids involved in the intersection are solids or voids



Solid with solid



Solid with void



Void with void

Consider: Are the shapes of the intersection lines the same for the above three forms?

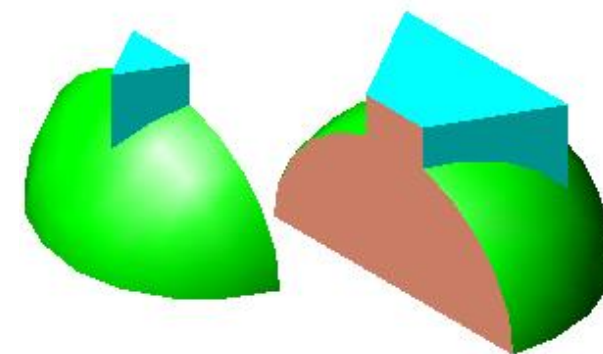
III. Methods for Finding Intersection Lines

1. The essence of finding the intersection line is to find a series of **common points** on the surfaces of the two intersecting solids.

Find special position points and general position points, and then connect these points with a curve to form the intersection line.

2. The basic method for finding the intersection line is **the auxiliary plane method**.

The drawing principle is to use **the common point of three surfaces**.



3. Principles for selecting the auxiliary plane:

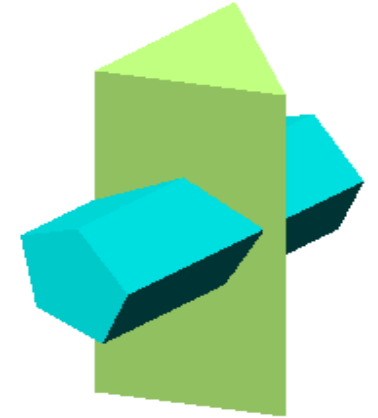
- The auxiliary plane is **parallel to the projection plane** (choose one).
- Make the intersection lines between the auxiliary plane and the two solids the simplest and easiest to draw, such as **straight lines or circles**.

5-2 Drawing of Intersection Lines



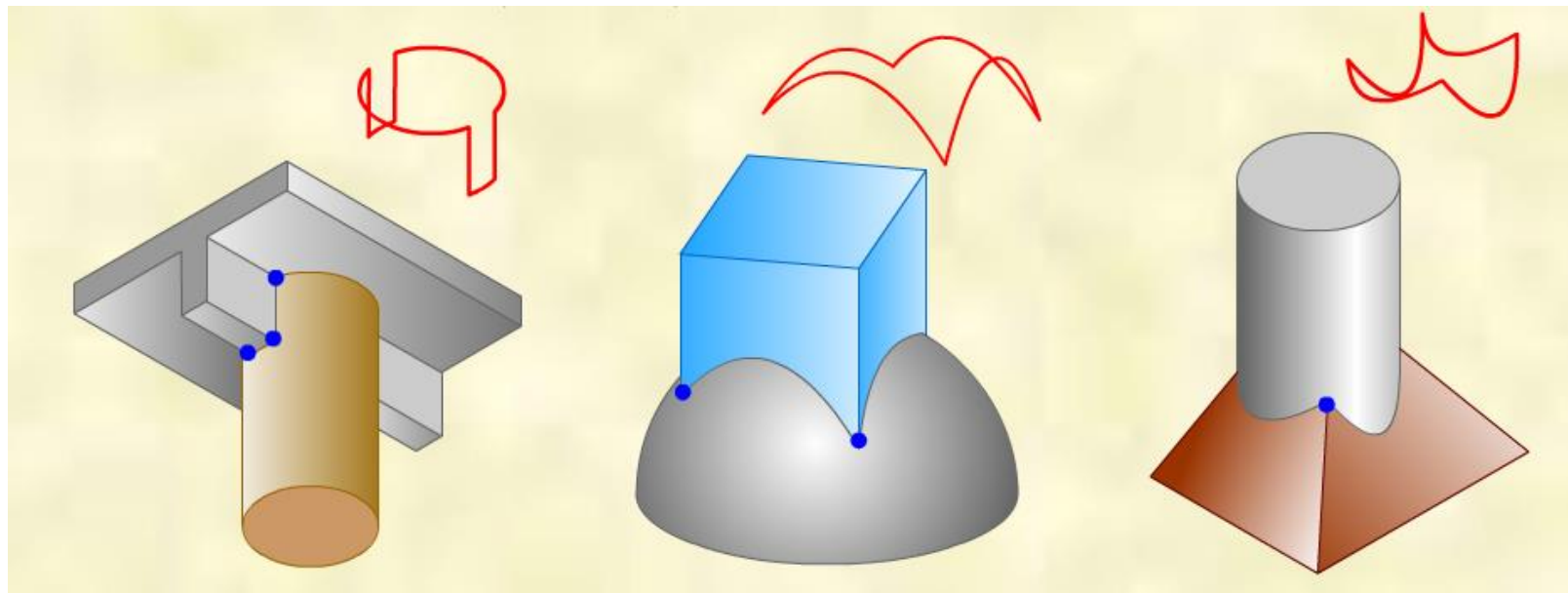
1. Intersection of Plane Solids

The intersection line is composed of several segments of plane straight lines.



2. Intersection of a Plane Solid and a Curved Solid

The intersection line is composed of several segments of plane curves.



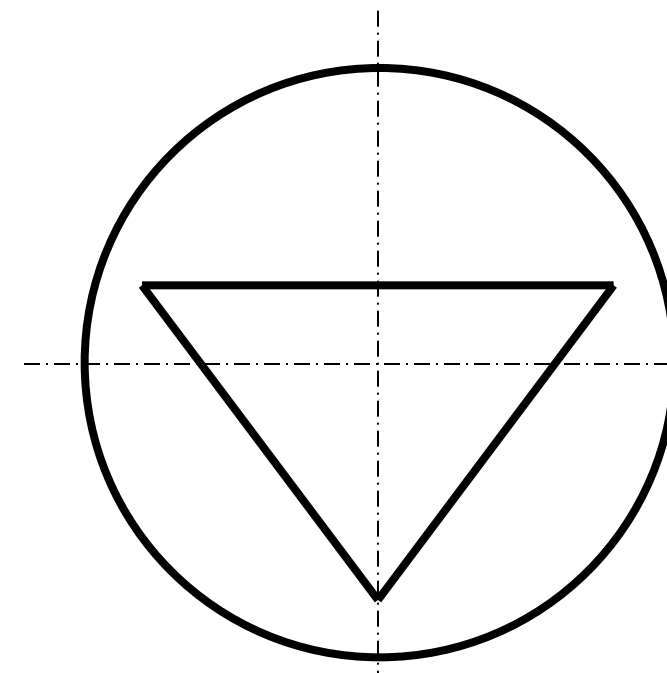
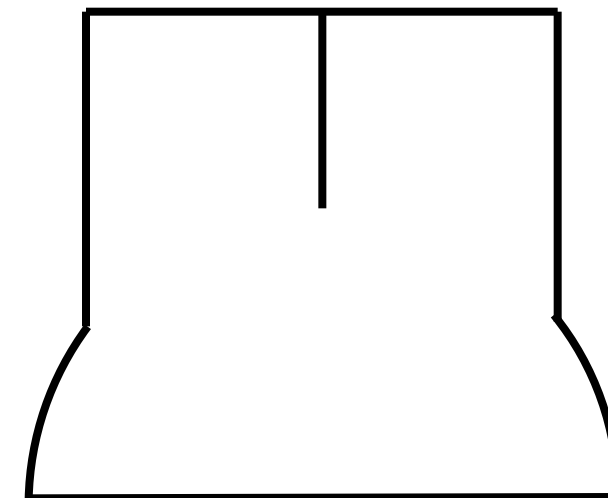
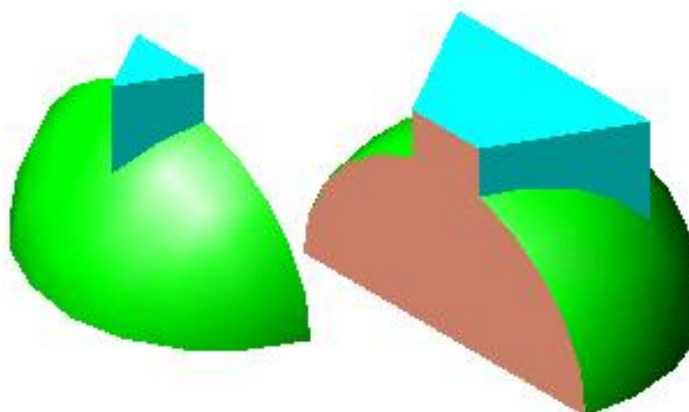
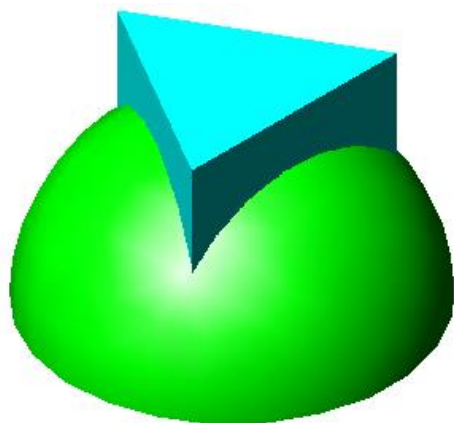
5-2 Drawing of Intersection Lines



Example: Find the intersection line between a right triangular prism and a hemisphere.

Solution Steps:

- 1) Shape Analysis 1: Identify the basic solids and their relative positions.
- 2) Shape Analysis 2: Surface properties and relative positions of the basic solids.
- 3) Analyze the known projections and range of the intersection line.
- 4) Choose an appropriate auxiliary plane: a plane parallel to the projection plane, with **the intersection line being a straight line or a circle.**



5-2 Drawing of Intersection Lines



5) Find the special points on the intersection line.

Frontmost, rearmost, leftmost, rightmost point

Boundary points of visibility and invisibility in each view:

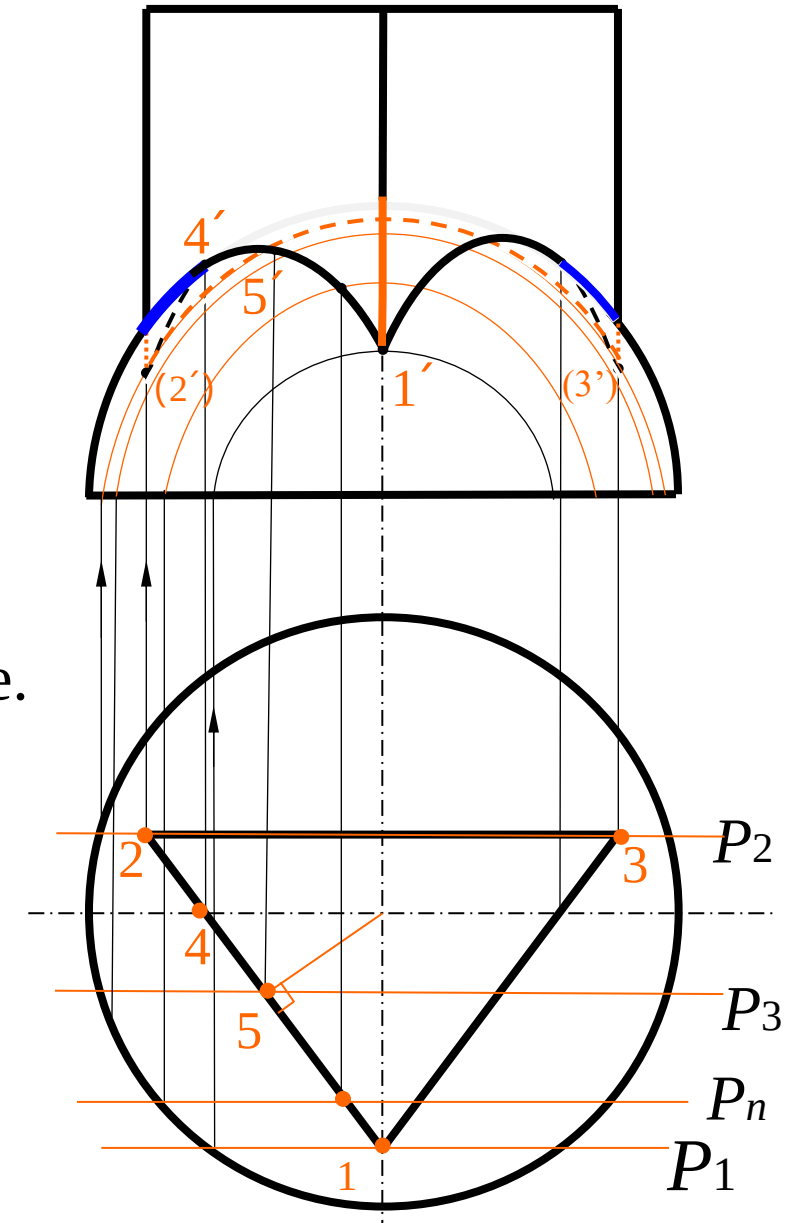
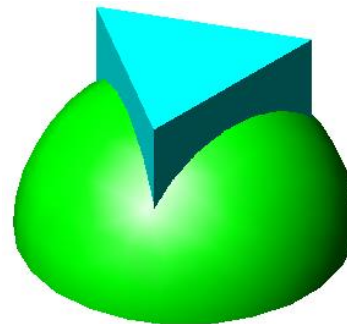
Highest point, Lowest point

6) Find a sufficient number of intermediate points.

7) Determine visibility and connect the points with a curve.

The intersection line is visible when both surfaces of the solids are visible.

8) Complete the contour lines.

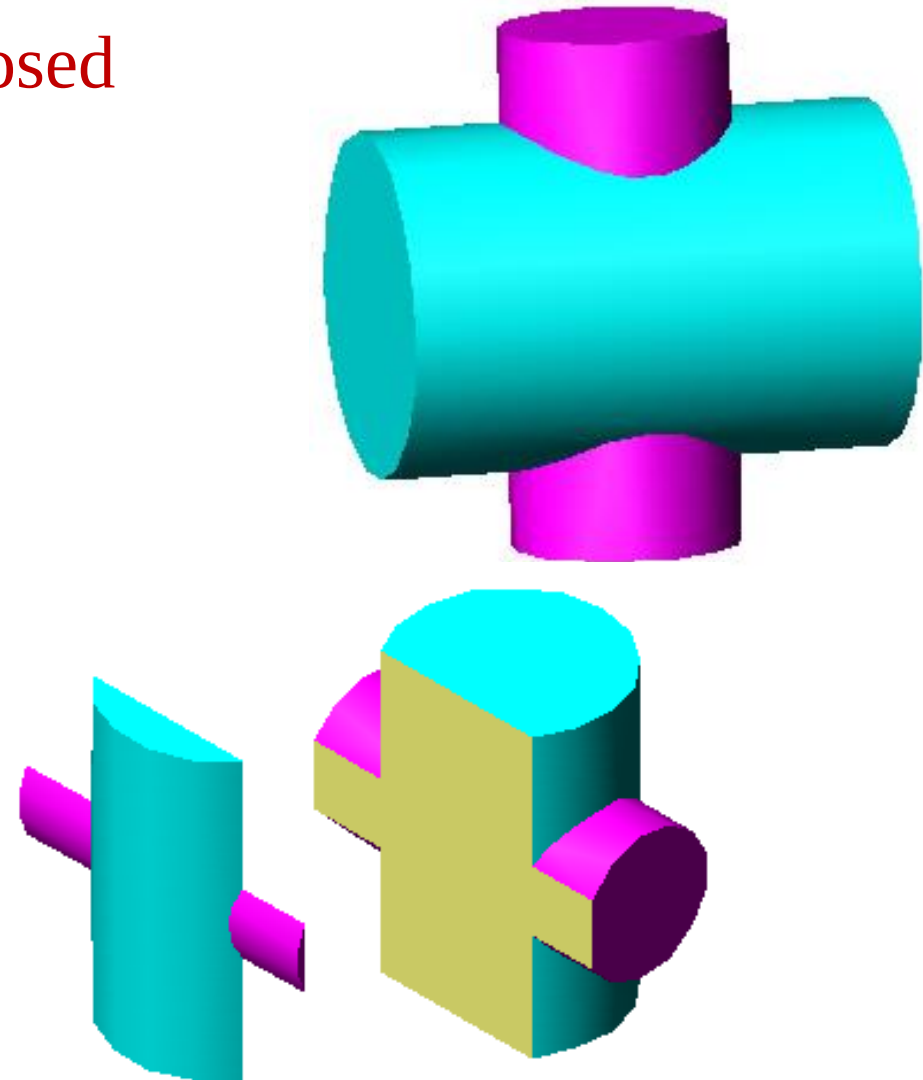
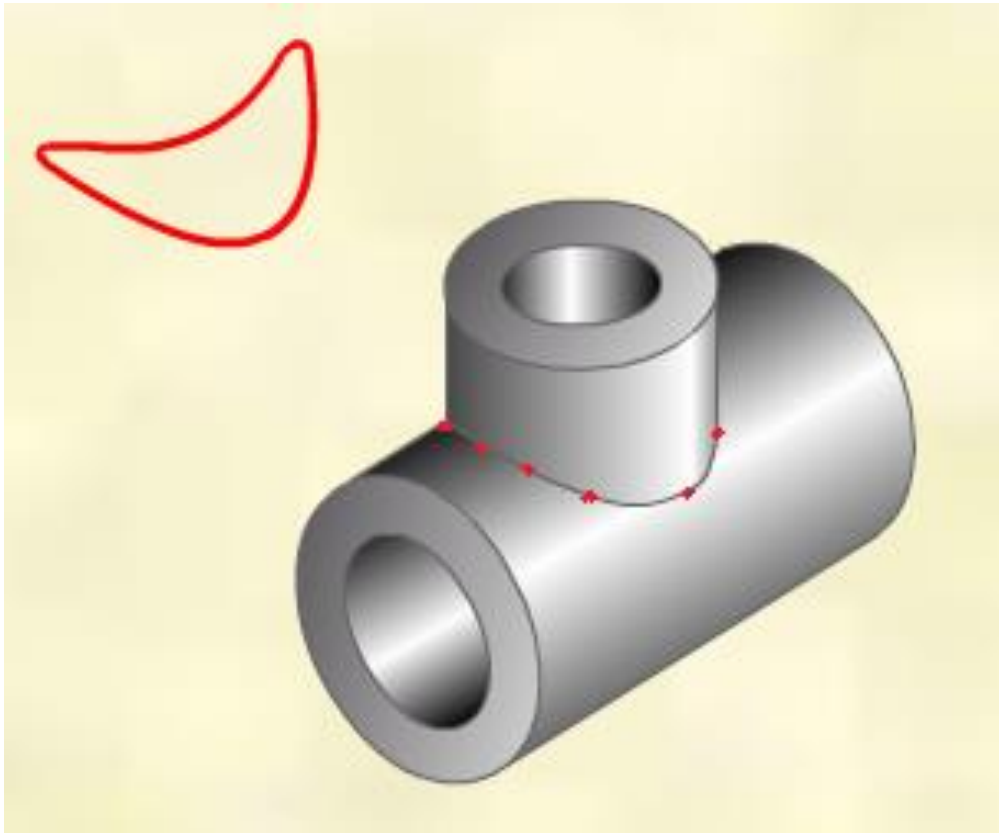


5-2 Drawing of Intersection Lines



3. Intersection of Two Solids of Revolution

The intersection line is generally a **closed** spatial curve.



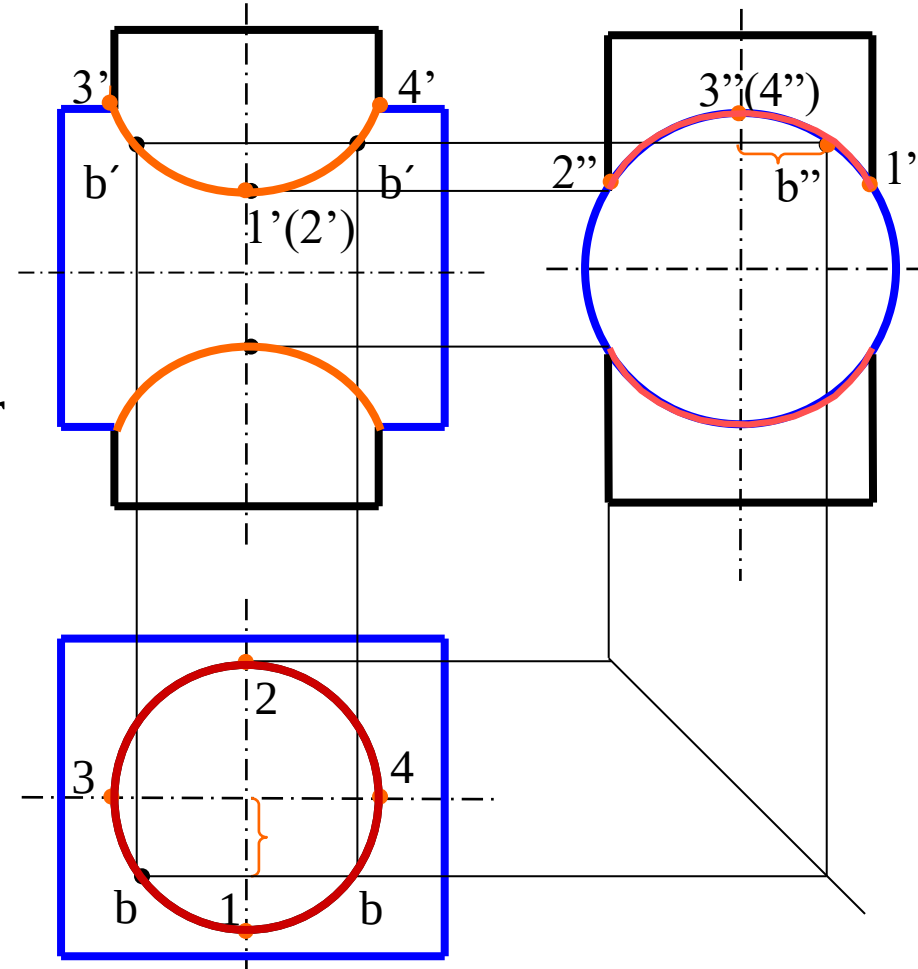
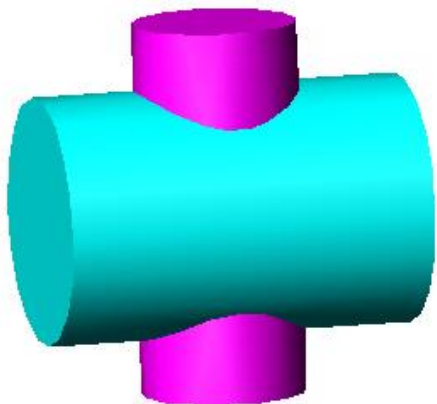
5-2 Drawing of Intersection Lines



Example: Find the intersection line.

Solution steps:

- 1) Analyze the shapes
- 2) Analyze the intersection line
- 3) Find special points
- 4) Find a series of intermediate points
- 5) Smoothly connect the points based on their visibility
- 6) Complete the contour lines



5-2 Drawing of Intersection Lines



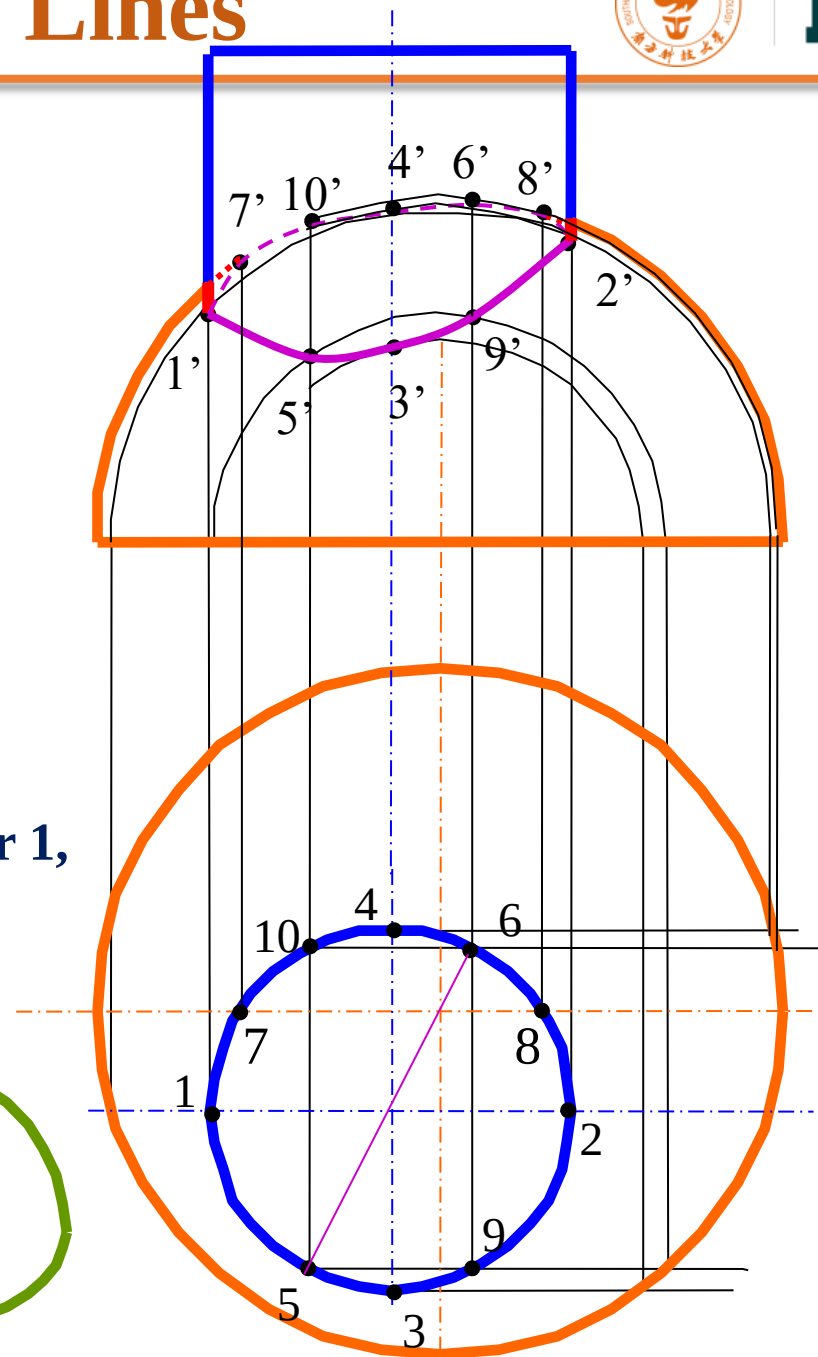
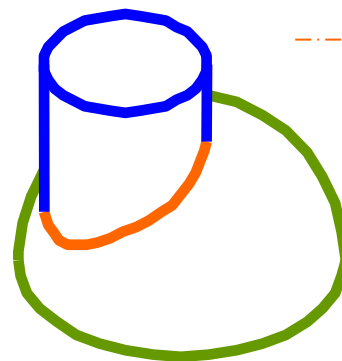
Example: Intersection of a cylinder and a hemisphere (not a key point)

Solution steps:

- 1) Shape analysis, determine the range of the intersection line
- 2) Choose an appropriate auxiliary plane
- 3) Find the special points on the intersection line
 - Leftmost point 1 Rightmost point 2
 - Frontmost point 3 Rearmost point 4
 - Highest point 6 Lowest point 5

Boundary points of visibility and invisibility 7, 8 or 1, 2?

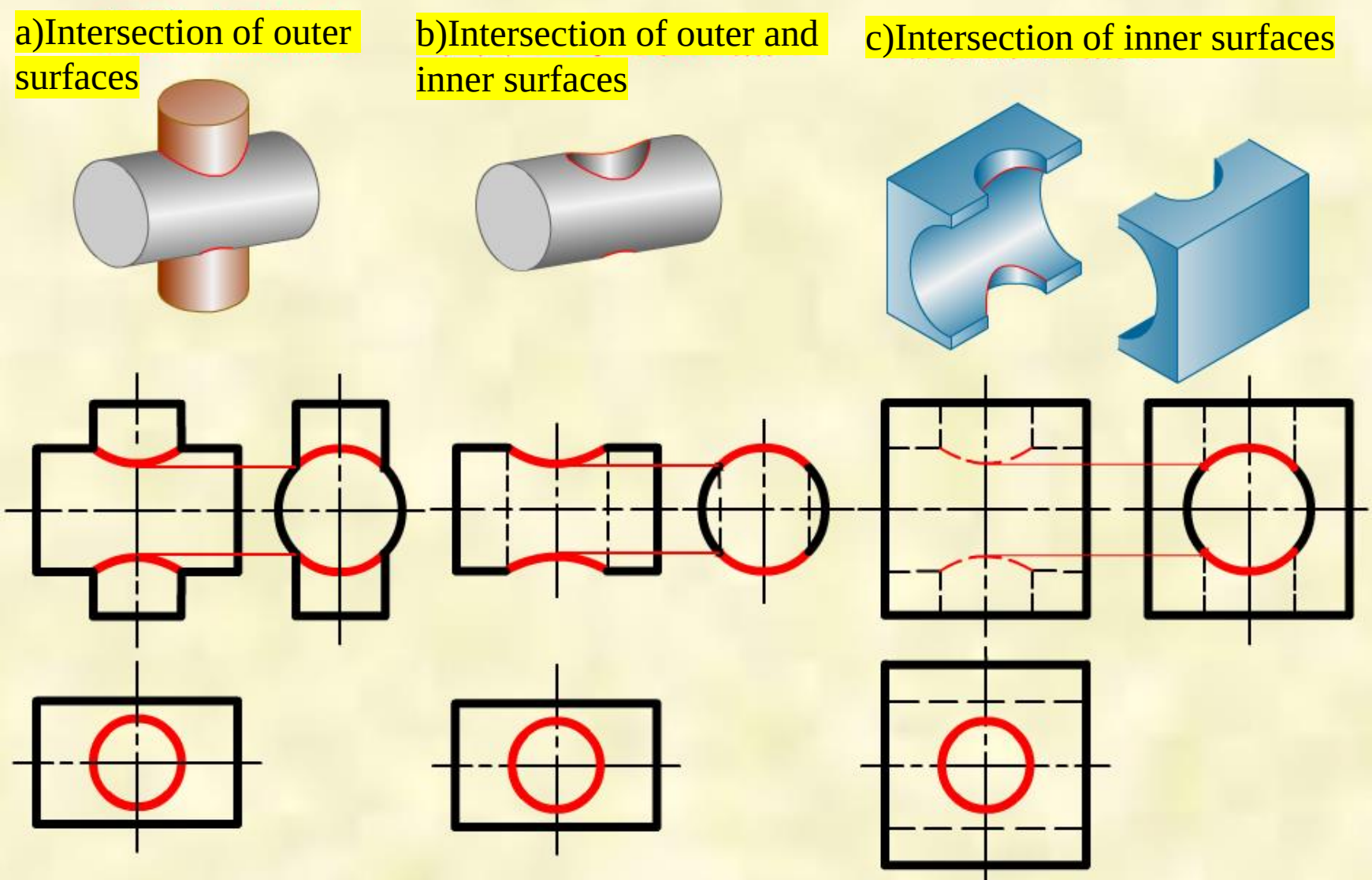
- 4) Find intermediate points
- 5) Smoothly connect the points based on their visibility
- 6) Complete the contour lines



5-2 Drawing of Intersection Lines



4. Different Forms of Intersection of Solids of Revolution



For two solids of revolution with identical dimensions, the shape of the intersection line is the same regardless of whether they are solid or void when intersecting each other.

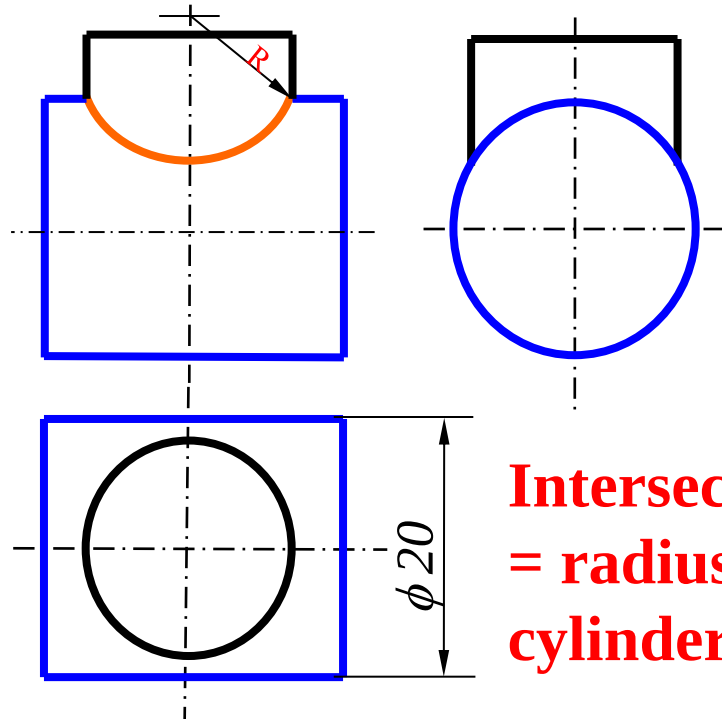
5-2 Drawing of Intersection Lines



5. Simplified Drawing Method for Intersection Lines

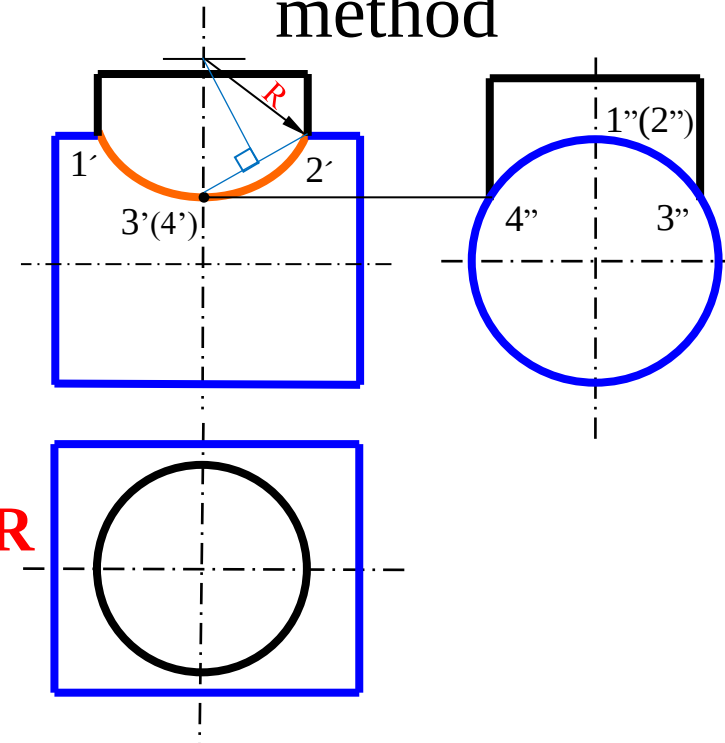
When high accuracy of the intersection line shape is not required, arcs can be used as a substitute.

- Large cylinder radius method



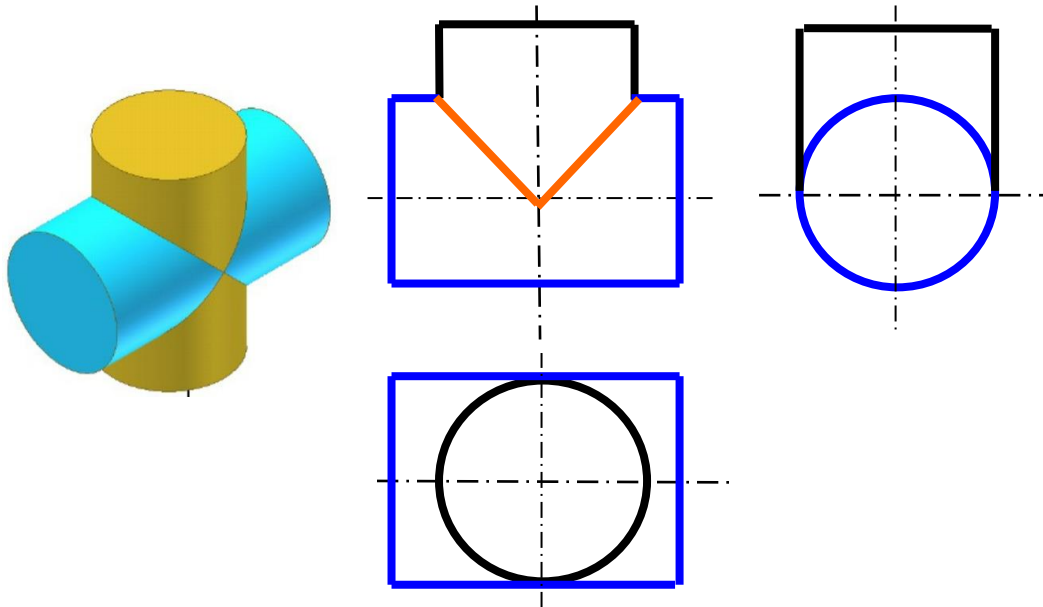
**Intersection line takes R
= radius of the large
cylinder**

- Median line method



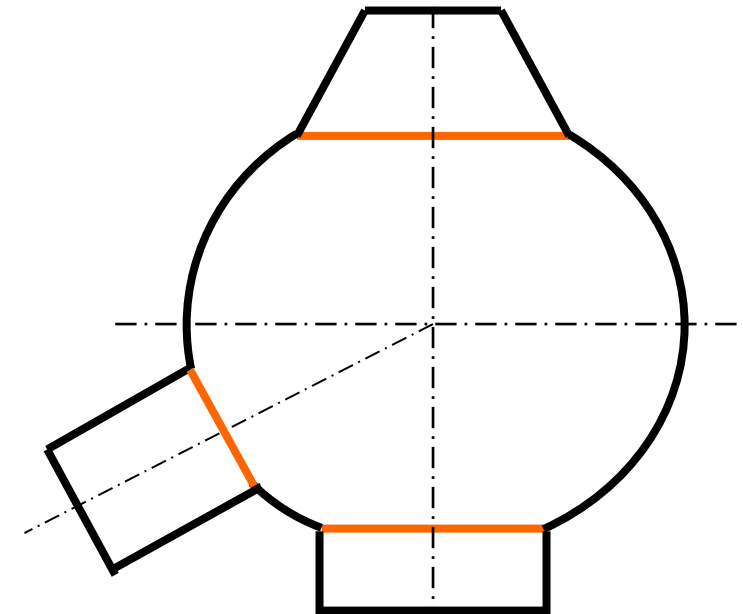
6. Special Cases of Intersection Lines

- Intersection of cylinders with equal diameters



Consider: From the perspective of the intersection plane, why is the intersection line a straight line?

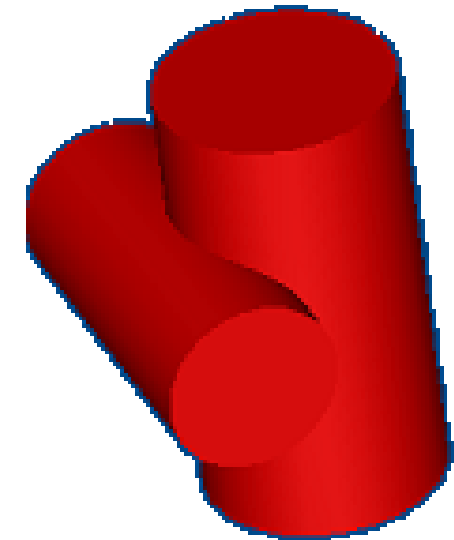
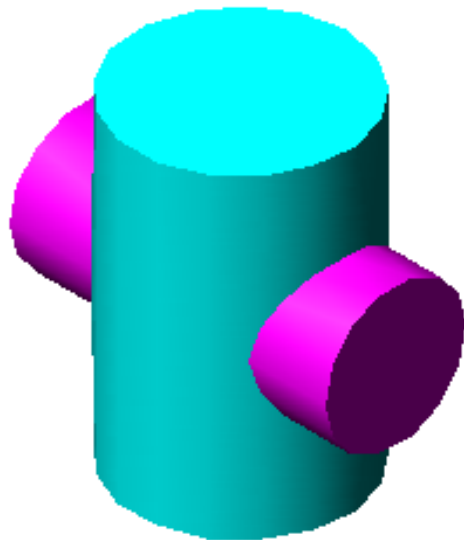
- Intersection of a sphere with a solid of revolution



The axis of the solid of revolution passes through the center of the sphere

7. Factors Affecting the Shape of Intersection Lines

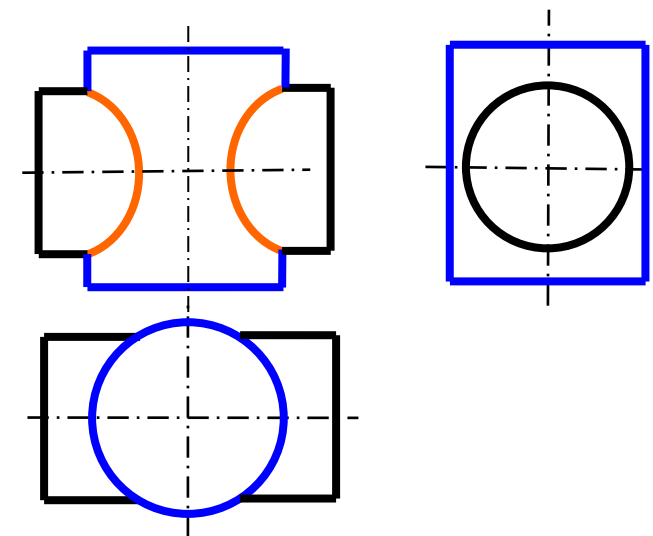
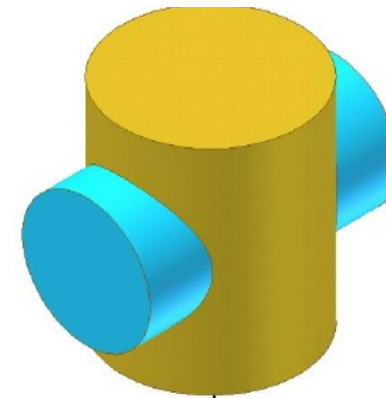
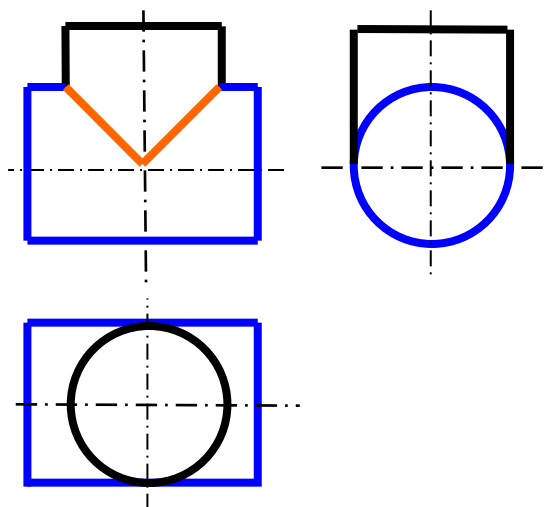
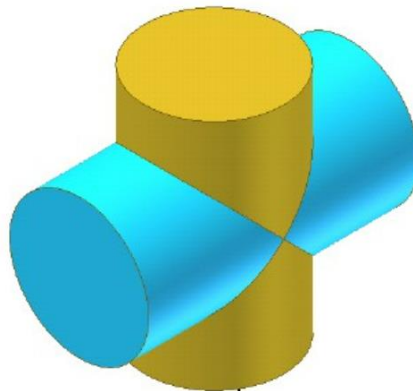
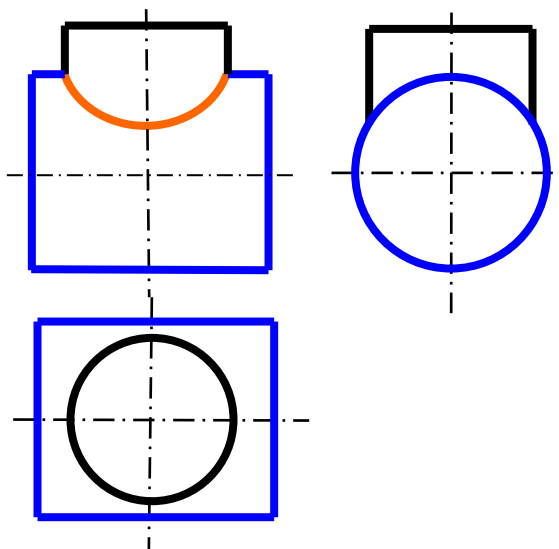
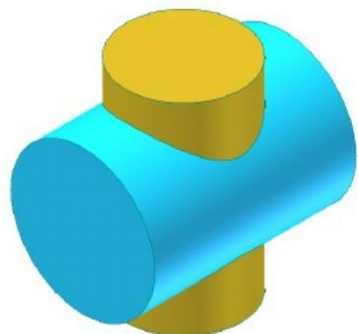
- (1) The nature of the surfaces of the two basic bodies involved in the intersection
- (2) Relative position



5-2 Drawing of Intersection Lines



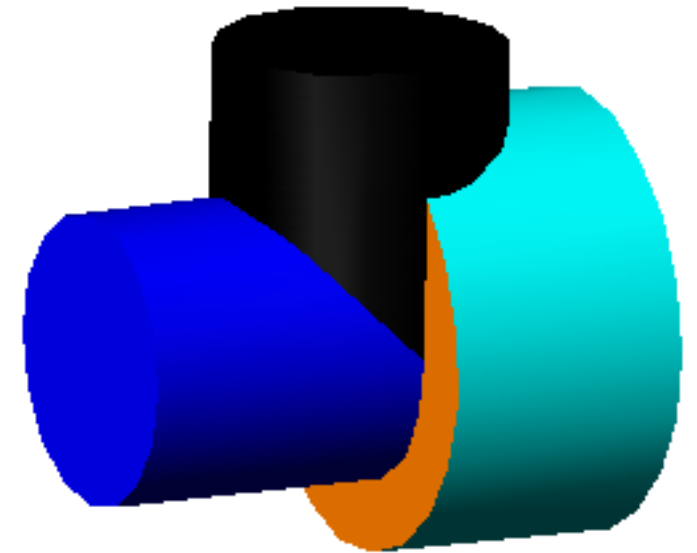
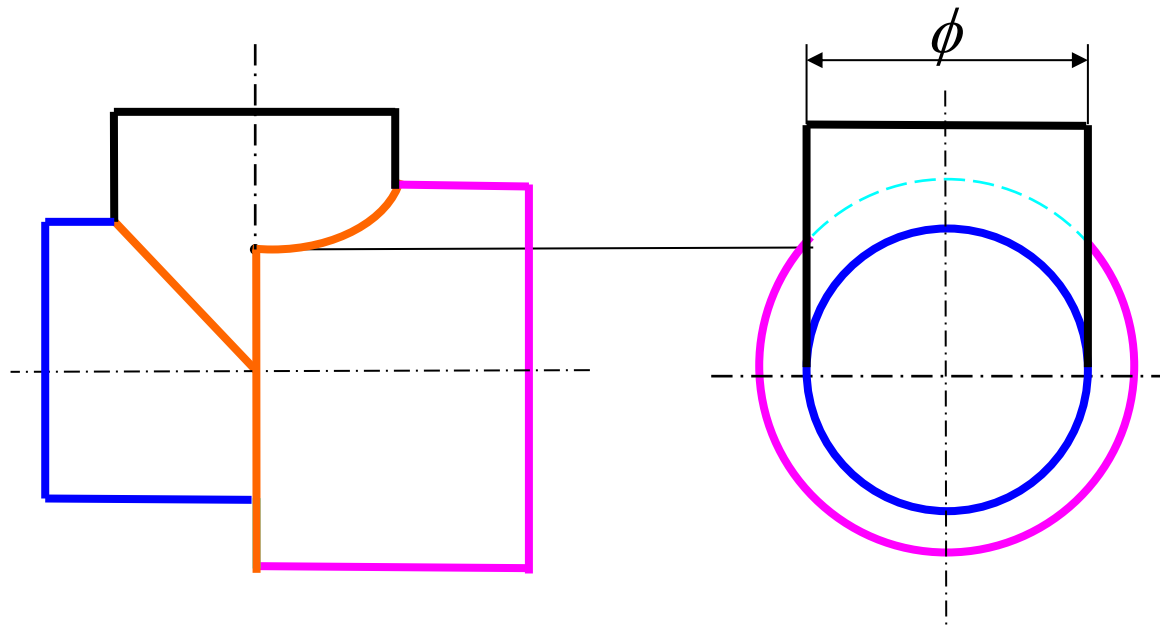
(3) Size



8. Combined Intersections

Intersection involving more than two solids of revolution

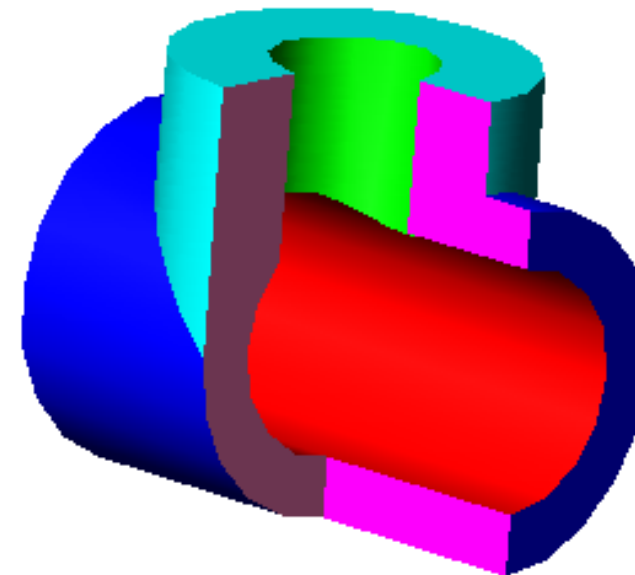
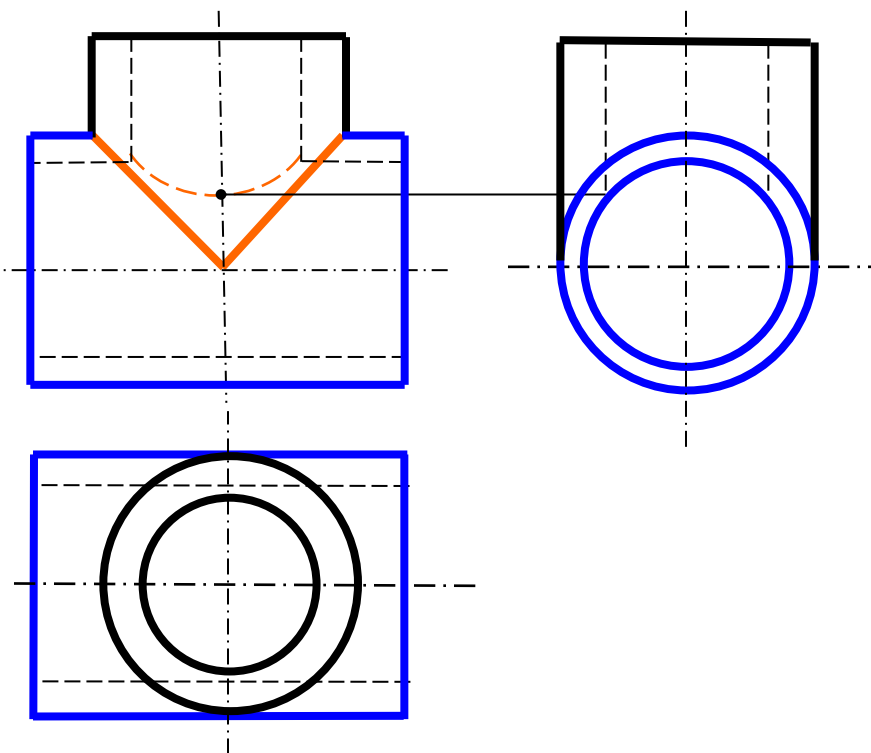
Example: Complete the front view



5-2 Drawing of Intersection Lines



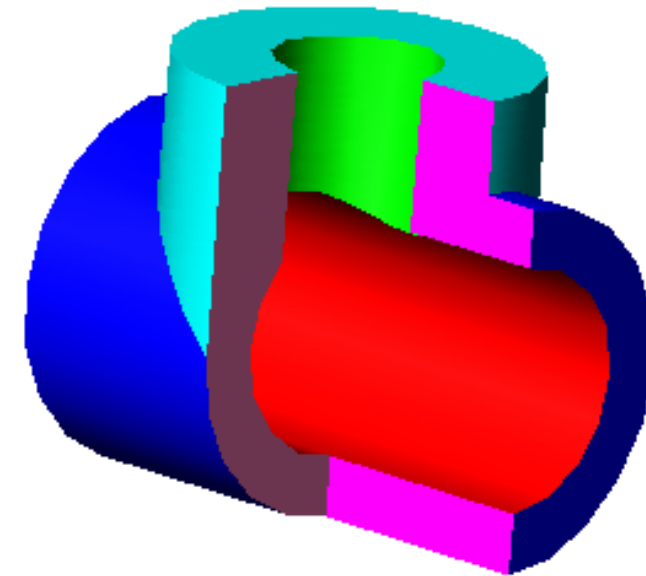
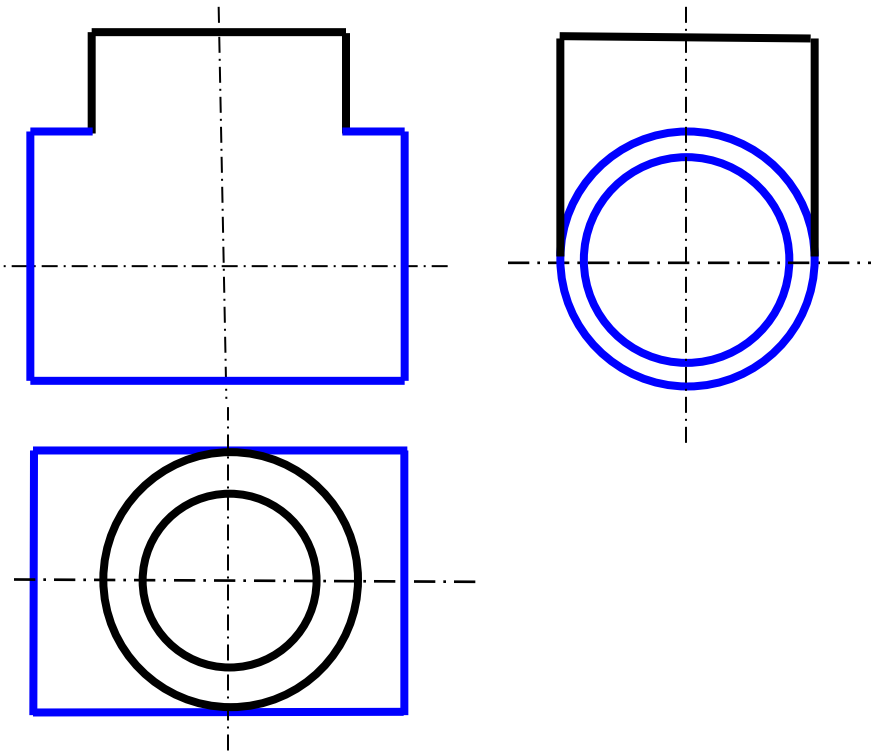
Example: Complete the front view



5-2 Drawing of Intersection Lines



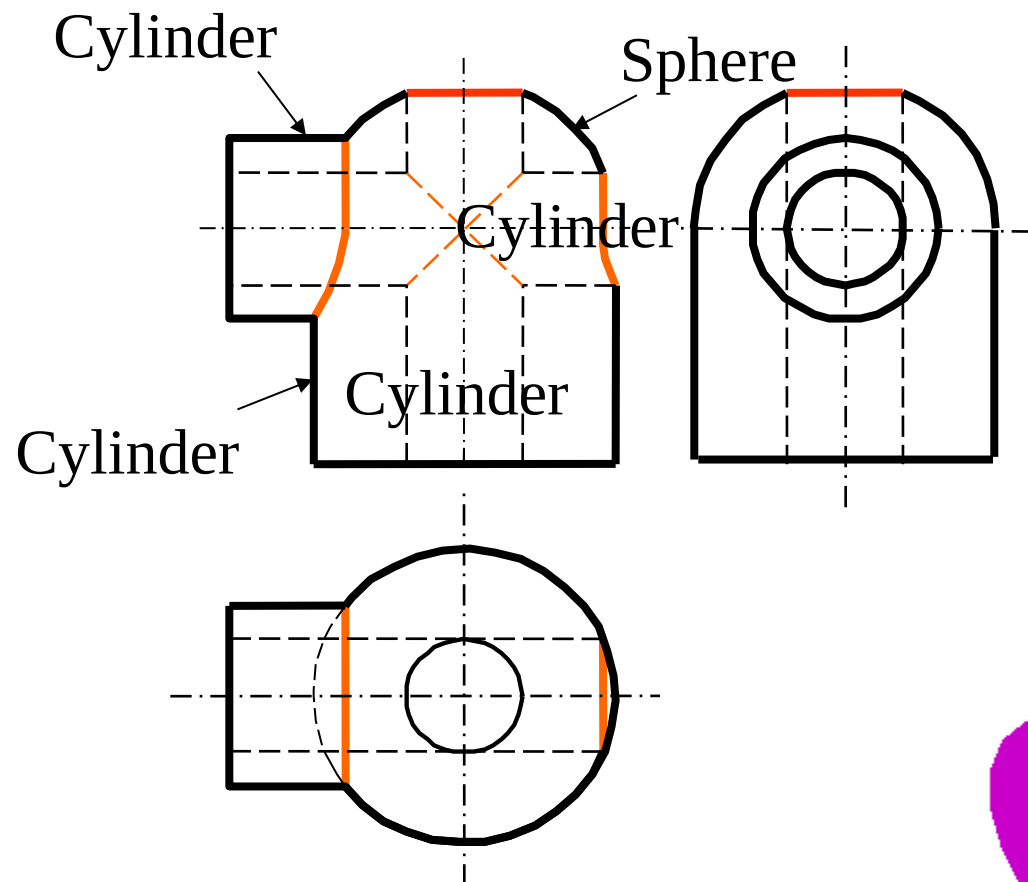
Example: Complete the front view



5-2 Drawing of Intersection Lines



Example: Add missing lines



1. Shape Analysis

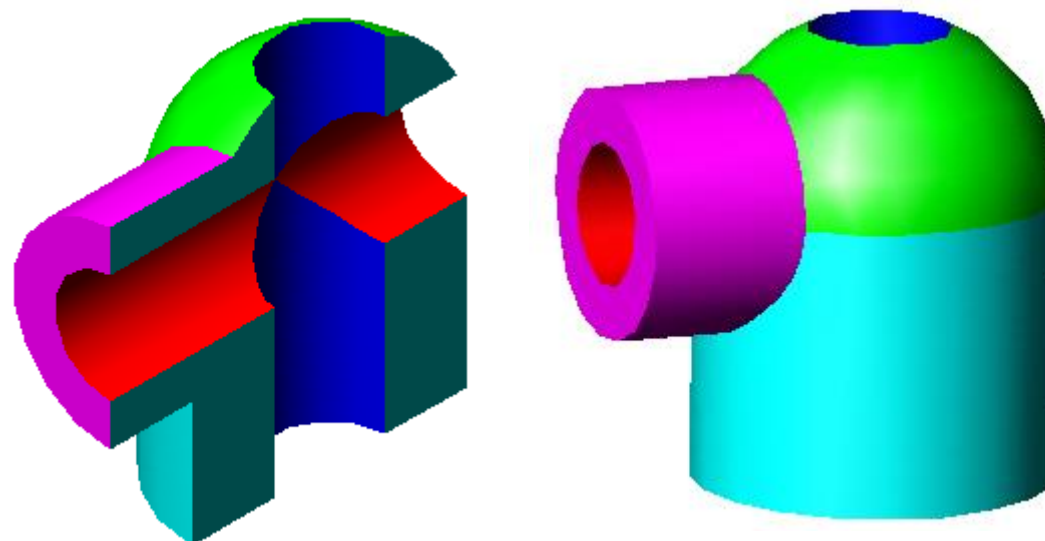
Outer surfaces: Cylinder, Sphere, Cylinder

Inner surfaces: Cylinder, Cylinder

2. Analyze the intersection of outer surfaces

3. Analyze the intersection of inner surfaces

4. Analyze the intersection of outer and inner surfaces





Practice problems

**Let's practice together, and I'll make sure
you understand! 😊**

Thank you !

Q&A

(Answer all questions)